

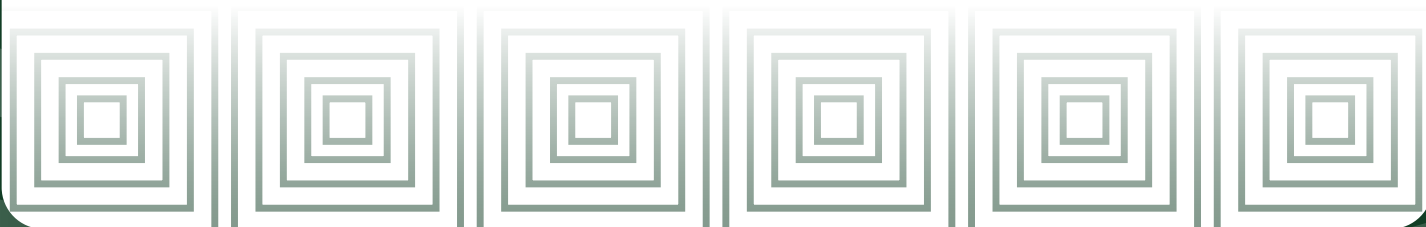
MATHEMATICS SYLLABUS

FOR ADVANCED LEVEL (S4-S6)

LEARNING PATHWAYS: MATHEMATICS AND SCIENCES

Second Edition

Kigali, 2026



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FOREWORD

The Rwanda Basic Education Board (REB) is honoured to present the second edition of the Upper Secondary School Mathematics syllabus (S4-S6) for Mathematics and Sciences Learning pathway. It serves as an official guide for competence-based teaching and learning, ensuring consistency in delivering quality education across the country. Developed collaboratively by subject specialists, educators, and curriculum developers, it supports the implementation of the Competence-Based Curriculum (CBC) and reflects Rwanda's commitment to nurturing students' knowledge, skills, values, and attitudes.

Quality education is the foundation of Rwanda's sustainable development and prosperity. Aligned with the national education philosophy, this syllabus promotes critical thinking, creativity, inclusiveness, and holistic learner development. It aims to empower young people with competencies necessary to achieve full potential, equipping them with relevant knowledge, skills, the right attitudes and values to integrate effectively into society and pursue opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the National Strategy for Transformation (NST2), this syllabus supports the nation's goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritizes learner-centred pedagogy, active learning, and the practical application of knowledge to prepare student not only for academic success but also to become responsible, ethical, and productive citizens capable of contributing to Rwanda's continued transformation.

This syllabus reflects our determination to equip students with 21st-century skills, foster innovation, and instil values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our students and supports.

Together, we can ensure that education continues to be a powerful driver of our nation's development and prosperity.



Dr. MBARUSHIMANA Nelson
Director General

ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this Upper Secondary school Mathematics syllabus (S4–S6) for Mathematics and Science Learning Pathway.

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda's Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda's education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among students. Through your dedication, this syllabus will fulfil its purpose of empowering students to realize their full potential and contribute meaningfully to Rwanda's sustainable development.



MURUNGI Joan

Head of Curriculum Teaching and Learning Resources Department/REB

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Experimental version

1. INTRODUCTION

1.1 Background to syllabus adaptation

The adaptation of the Mathematics syllabus for Upper Secondary school reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This second edition incorporates the Ministerial directive to harmonize school hours to five per day and 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as Vision 2050 and the National Strategy for Transformation (NST2), which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

The new upper secondary Mathematics syllabus guides the interaction between the teacher and the learner through the learning processes and highlights the essential practical skills and competences a student should acquire during and at the end of each unit of learning.

1.2 Rationale of teaching and learning mathematics

1.2.1 Mathematics and society

The Mathematics syllabus has put emphasis on integrated production skills and on an integrated approach in all disciplines. Mathematics is an excellent vehicle for the development and improvement of a person's intellectual competence in logical reasoning, spatial visualization, analysis and abstract thought. Learning mathematics develops numeracy, logical reasoning skills, critical thinking skills and problem-solving skills. This will result in mathematics being used in many activities of daily life thereby serving as an important tool to the society. In this way the subject will be demystified and user friendly.

Therefore, mathematics plays an important role in society through abstraction and logic, counting, calculation, measurement, systematic study of shapes and motion. It is also used in natural sciences, engineering, medicine, finance and social sciences. Applied Mathematics like statistics and probability play an important role in game theory, in the national census process, in scientific research, etc. In addition, some cross-cutting issues such as financial awareness are incorporated into some of the mathematics units to improve the social and economic welfare of Rwandan society.

1.2.2 Mathematics and students

Students need enough basic mathematical competences to be effective members of Rwandan society, including the ability to count, estimate, measure, calculate, handle and manage money, interpret statistics, assess probabilities, and read commonly used mathematical representations and graphs. Reading or listening to the news requires many of these competences and citizenship requires being able to interpret critically the information one receives. For example, understanding an age-length or age-weight graph helps parents and health practitioners monitor the health of a child.

Mathematics also equips students with knowledge, skills and attitudes necessary to enable them to succeed in an era of rapid technological growth and socio-economic development. Mastery of basic mathematical ideas and operations should make students confident in problem-solving in life situations. A high-quality mathematics education therefore provides a foundation for understanding the world, the ability to reason mathematically, an appreciation of the beauty and power of mathematics, and a sense of enjoyment and curiosity about Mathematics subject.

In this syllabus the teacher has the task of trying to make mathematics a reality in life. Methods and approaches to learning experiences should be mostly practical and based on the experience of the students. Hence, teaching methods to be emphasized are those that allow students to explore, try different procedures and solve problems practically. Learning mathematics needs to include practical problem-solving activities with opportunities for students to plan their own investigations and develop their mathematical competency and confidence.

Moreover, as new technologies have had a dramatic impact on all aspects of life, wherever possible in mathematics, students should gain experience of a range of ICT applications.

1.2.3 Competences

Competence is defined as the ability to perform a particular task successfully, resulting from having gained an appropriate combination of knowledge, skills and attitudes.

The mathematics syllabus provides the opportunity for students to develop different competencies, including the generic competencies.

Basic competencies are addressed in the stated broad subject competences and in objectives highlighted year on year basis and in each unit of learning. The generic competencies, basic competences that must be emphasized and reflected in the learning process are briefly described below and teachers will ensure that students are exposed to tasks that help the students acquire the skills.

a) Generic competences and values

- **Critical and problem-solving skills:** Students use different techniques to solve mathematical problems related to real life situations. They are engaged in mathematical thinking, they construct, symbolize, apply, and generalize ideas.

The acquisition of such skills will help students to think imaginatively and broadly to evaluate and find solutions to problems encountered in all situations.

- **Creativity and innovation:** The acquisition of such skills will help students to take initiatives and use imagination beyond the knowledge provided to generate new ideas and construct new concepts. Students will improve these skills through mathematics contests, and mathematics competitions, etc.
- **Research:** This will help students find answers to questions based on existing information and concepts as well as explain phenomena based on findings from information gathered.

- **Communication in official languages:** Students communicate effectively their findings through explanations, construction of arguments and drawing relevant conclusions.

Mathematics teachers, irrespective of not being teachers of language, will ensure the proper use of the language of instruction by students. This will help students communicate clearly and confidently and convey ideas effectively through speaking and writing and use the correct language structure and relevant vocabulary.

- **Cooperation, interpersonal management and life skills:** Students are engaged in cooperative learning groups to promote higher achievement rather than competitive and individual work.

This will help students to co-operate with others as a team in whatever task are assigned and to practice positive ethical moral values and respect for the rights, feelings and views of others. Learners will perform practical activities related to environmental conservation and protection. They will also advocate for personal, family and community health, hygiene and nutrition and respond creatively to the variety of challenges encountered in life.

- **Lifelong learning:** The acquisition of such skills will help students update their knowledge and skills with minimum external support and to cope with the evolution of advances in knowledge for personal fulfilment in areas that need improvement and development.

b) Broad mathematics competences

Upon completion of the S4-S6 cycle, the student will be able to:

- Construct and present logical arguments with clarity, coherence, and precision, both verbally and in writing.
- Master advanced mathematical concepts and apply them accurately to solve real life problems.
- Analyze and apply mathematical models to understand and interpret real-world phenomena.
- Systematically collect, organize, analyze, and interpret data to inform conclusions and support decision-making.

- Demonstrate intellectual curiosity and a research-oriented approach to mathematical theories and their applications.
- Utilize ICT tools including AI, to explore mathematical concepts and solve problems.
- Cultivate professional competencies such as self-directed work, effective teamwork, and efficient time management.
- Show a strong foundation for specialized further studies in mathematics, sciences, and related fields at the tertiary level.

c) Mathematics and developing competences

The national policy documents based on national aspirations identify some 'Basic Competencies' alongside the 'Generic Competencies' that will develop higher order thinking skills and help students learn subject content and promote the application of acquired knowledge and skills.

Through observations, constructions, hands-on manipulations, generalizations, and presentations of information during the learning process, the learner will not only develop deductive and inductive skills but also acquire co-operation, communication, critical thinking and problem-solving skills. This will be realized when students make presentations leading to inferences and conclusions at the end of the learning unit. This will be achieved through group work activities and co-operative learning which in turn will promote interpersonal relations and teamwork.

The acquired knowledge in learning mathematics should develop a responsible citizen who adapts to scientific reasoning and attitudes and develops confidence in reasoning independently.

1.3 Pedagogical approach

The change to a competence-based curriculum is about transforming learning to ensure that learning is deep, enjoyable and habit-forming.

Various teaching strategies and approaches such as direct instruction, discovery learning, investigation, guided discovery or other methods must be incorporated. Among the approaches that can be given consideration include the following:

- Learner-cantered learning.
- Different learning abilities and styles of students (individualization);
- Co-operative learning.
- Contextual learning.
- Mastery learning.
- Constructivism.
- Use of relevant, suitable and effective teaching materials.
- Formative evaluation to determine the effectiveness of teaching and learning processes.

In addition, IT in general and particularly ICT and AI should be used as a pedagogical tool to facilitate teaching and learning of mathematics.

The choice of a suitable approach will stimulate the teaching and learning environment inside or outside the classroom.

1.3.1 The role of the learner

In the competence-based curriculum, the learner is the principal actor of his/her education. He/she is not an empty bottle to fill. Taking into account the initial capacities and abilities of the learner, the activities of the learner are indicated against each learning unit and reflect appropriate engagement of the learner in the learning process. The teaching-learning process will be tailored towards creating a learner friendly environment based on capabilities, needs, experience and interests.

Therefore, the following are some of the roles or expectations from students:

- Students construct the knowledge either individually or in groups in an active way. From the learning theory, students move in their understanding from concrete through to pictorial to abstract. Therefore, the opportunities should be given to students to manipulate concrete objects and to use models.
- Students will be encouraged to do research and present their findings through group work activities.
- A learner is co-operative: students work in heterogeneous groups to increase tolerance and understanding.
- Students are responsible for their own participation and for making sure others participate.
- Help is sought from within the group and the teacher is asked for help only when the whole group agrees to ask a question.
- Consensus on the answer is required from the whole group.
- The group evaluates its own strategies and ideas rather than relying on the teacher for this evaluation.
- The students who learn at a faster pace do not do the task alone and then the others merely sign off on it.
- Participants ensure the effective contribution of each member, through clear explanation and articulation of constructive arguments, to improve their English literacy, develop a sense of responsibility and to increase their self-confidence, and public speaking ability, etc.

1.3.2 The role of the teacher

Some of the specific duties of the teacher when implementing competence-based activities are as follows:

- He/she is a facilitator: his/her role is to provide opportunities for students to meet problems that create interest and challenge them and that, with appropriate effort, they can solve.

- He/she is an organizer: his/her role is to organize the students, in the classroom or outside, and engage them through participatory and interactive methods through the learning processes as individuals, in pairs or in groups. To ensure the learning is personalized and active and participative, the teacher must identify the needs of the students, the nature of the learning to be done, and the means to shape learning experiences accordingly.
- He/she is an advisor: he/she provides counselling and guidance for students in need. He/she comforts and encourages students by valuing their contributions in the class activities.
- He/she is a conflict-solver: when members of a group have problems such as the attribution of tasks, he/she should provide useful and constructive ideas. The teacher should settle disputes among the group.
- He/she is ethical: he/she teaches by example, by being impartial, by being a role-model, and by caring for individual needs, especially for slow students and students with special educational needs.

1.3.3 Special needs education and inclusive approach

All Rwandans have the right to access education regardless of their different needs. The underpinnings of this provision would naturally hold that all citizens benefit from the same menu of educational programs. The possibility of this assumption is the focus of special needs education. The critical issue is that we have persons/students who are totally different in their ways of living and learning as opposed to the majority. The difference can either be emotional, physical, sensory and/or intellectual learning challenged, traditionally known as mental retardation.

These students equally have the right to benefit from the free and compulsory basic education in nearby ordinary/mainstream schools. Therefore, the schools' role is to enrol them and also set strategies to provide relevant education for them. The teacher therefore is requested to consider each learner's needs during the teaching and learning process. Assessment strategies and conditions should also be standardized to the needs of these students. Detailed guidance for each category of students with special education needs is provided for in the guidance for teachers.

1.4 Assessment approach

Assessment evaluates the teaching and learning process through collecting and interpreting evidence of an individual learner's learning progress and makes a judgment about the learner's achievements measured against defined standards. Assessment is an integral part of the teaching learning process. In the new competence-based curriculum, assessment must also be competence-based, whereby a learner is given a complex situation related to his/her everyday life and asked to try to overcome the situation by applying what he/she has learned.

Assessment will be organized at the following levels: School-based assessment, District examinations, National assessment Learning Achievement in Rwandan Schools (LARS), and National examinations.

1.4.1 Levels of assessment in Advanced Level across terms and national examinations

To provide a comprehensive picture of learner achievement and development, assessment is conducted at multiple levels:

- i) **Diagnostic:** Conducted before instruction to identify students' prior knowledge and skills; administered at the beginning of a unit or lesson to guide teaching.
- ii) **Formative / Continuous:** Carried out during the learning process to track progress and provide feedback; implemented continuously through quizzes, projects, class activities, and teacher observations.
- iii) **Summative assessment:**
 - **End Term I:** Administered after instruction to evaluate students' achievement of Term I content; conducted at the end of the first term as a school-based assessment.
 - **End Term II:** District-coordinated assessment designed to measure achievement of Term II content; developed by schools and quality-checked at district level.
 - **End Term III:** National assessment to evaluate mastery of the entire year's content; prepared, set, and moderated by NESA at the end of the third term.

- **National Examinations** : High-stakes assessments for certification, selection, and accountability; administered nationally at the end of key education cycles (P6, S3, S6). This examination integrates objective questions, structured responses, and applied tasks aligned with CBC principles to holistically measure learner achievement.
- In Advanced Level, all students will take 11 subjects in total -7 core subjects and 4 compulsory subjects. The compulsory subjects include ICT, Entrepreneurship, General and Religious Studies, and Physical Education and Sport, which are mandatory for all students across all pathways.
- All subjects will be assessed at the national level, except for Physical Education and Sport, General and Religious Studies, ICT, and Entrepreneurship, which will be assessed at the school level only.

Average scores from school, district, and comprehensive summative assessments will be weighted and included in the final national examination grading, initially contributing 10% with a percentage that can increase gradually.

1.4.2 Record keeping

Accurate record keeping is essential for monitoring learner progress and informing instructional decisions. Teachers use tools such as portfolios, checklists, and report books to collect evidence of learning. Portfolios include samples of students' work, test results, projects, and self-reflections, serving as both a record of achievement and a diagnostic tool for guiding remedial action. These records provide data for feedback to students, teachers, and parents, and contribute to final evaluations.

1.4.3 Item writing in summative assessment

Summative assessment items must be valid, fair, reliable, and aligned with CBC principles. They should test a range of competencies, including knowledge recall, application, analysis, and values integration, while reflecting real-life contexts. Item writers follow a structured process that involves:

- Identify topic areas to be tested from the subject syllabus.
- Outline subject matter content to be considered as the basis for the test.
- Identify learning outcomes to be measured by the test.
- Preparing a test blueprint or table of specifications;
- Using verbs that require higher-order thinking beyond memorization;
- Applying transparent and consistent marking schemes.
- Greater weight is given to higher-order cognitive skills as stated in the syllabus.

1.4.4 Structure and format of national examinations

There will be one paper in Mathematics at the end of upper secondary level. The paper will be composed of two sections, where the first section will be composed with short answer items or items with short calculations which include the questions testing for knowledge and understanding, investigation of patterns, quick calculations and applications of Mathematics in real life situations. The second section will be composed with long answer items or answers with demonstrations, constructions, high level reasoning, analysis, interpretation and drawing conclusions, investigation of patterns and generalisation. The items for the second section will emphasise the mastering of mathematical facts, the understanding of mathematical concepts and its applications in real life situations. In this section, the assessment will find out not only what skills and facts have been mastered, but also how well students understand the process of solving a mathematical problem and whether they can link the application of what they have learned to the context or to the real-life situation. The time required for the paper is at least three hours (3hrs).

The following topic areas must be assessed: Algebra, calculus, trigonometry, linear algebra, geometry, statistics and probability. Topic areas with more weight will have more emphasis in the second section where students should have the right to choose to answer 3 items out of 5.

1.4.5 Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent-teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where students are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well, meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports students' continued growth.

1.5. Resources

1.5.1 Materials needed for implementation

The following list shows the main materials and equipment needed in the learning and teaching process:

- Materials to encourage group work activities and presentations: Computers (desktops and laptops) and projectors; manila papers and markers
- Materials for drawing and measuring geometrical figures/shapes and graphs: Geometric instruments, ICT tools such as GeoGebra, Microsoft student ENCARTA.
- Materials for enhancing research skills: Textbooks and internet (the list of the textbooks to consult is given in the reference at the end of the syllabus and those books can be found in printed or digital copies)
- Materials to encourage the development of mathematical models: scientific calculators, Math type, MATLAB, etc.

Note that the technology used in the teaching and learning of Mathematics should be regarded as tools to enhance the teaching and learning process and not to replace teachers.

1.5.2 Use of AI in Mathematics teaching and learning

AI became a powerful ally in mathematics education when used thoughtfully. The key is viewing AI as a powerful tool in our educational toolkit, not as a threat to genuine mathematical understanding.

Here is how teachers and students can ask a question to AI at different stages of the learning process.

1. Ways in which Mathematics teachers can use AI

a) Lesson planning and content creation

- Generate practice activities: "Create 10 problems on quadratic equations with varying difficulty levels".
- Develop worksheets: "Make a worksheet on Pythagorean theorem with real-world applications".
- Create alternative explanations: "Explain complex numbers in 3 different ways for visual, auditory, and kinesthetic students".
- Generate assessment items: "Create a quiz on derivatives with answer key and scoring rubric".

b) Differentiation and individualization

- Adapt problems: "Modify this calculus problem for a student who's struggling with algebraic manipulation".
- Create assignments: "Provide this geometry proof at basic, proficient, and advanced levels".
- Generate enrichment: "Create challenging problems about geometric sequences for advanced students".

c) Assessment and feedback

- Analyze common errors: "Identify common misconceptions in solving linear equations and suggest addressing strategies".
- Generate rubrics: "Create a grading rubric for the activities on quantifiers and their negation".
- Provide feedback templates: "Suggest specific, constructive feedback for a student who made a sign error in solving inequalities".

d) Professional development

- Stay current: "What are recent research-based strategies for teaching fractions?"
- Plan interventions: "Suggest interventions for students struggling with word problems".
- Designing projects to be assigned to students: "Help me design a project-based learning unit on exponential growth".

2. Ways in which Mathematics students can use AI

a) Personalized tutoring and explanation

- Get multiple explanations: "Explain the chain rule in calculus using simple expressions to be understood by a S4 student".
- Clarify concepts: "Why do we 'invert and multiply' when dividing fractions? Explain visually".
- Step-by-step guidance: "Walk me through solving this simultaneous equations step by step using matrices, but don't give me the answer".

b) Practice and problem-solving

- Generate similar problems: "Give me more problems like this one about probability".
- Check understanding: "Create a self-check quiz on trigonometric identities".
- Explore variations: "What if this geometry problem had different given information?"

c) Conceptual deepening

- Make connections: "How is the concept of slope related to real-world applications?"
- Explore why: "Why does the quadratic formula work? Derive it for me".
- Historical context: "What's the history behind the development of calculus?"

3. Best practices and guidelines

For Teachers, use AI as a Co-planner, not replacement. Verify and edit AI-generated content for accuracy. Teach students HOW to use AI responsibly. Create AI-use policies for your classroom so that they focus on process over answers.

For Students, use AI as a tutor, not an answer key. Show your work and ask for feedback on process. Ask “why” and “how” not just “what’s the answer”. Do not copy AI solutions directly but use AI to understand methods. Do not depend solely on AI but use AI to supplement learning. Do not trust AI blindly but verify answers and methods.

In conclusion, the most effective mathematics classrooms will be those where teachers use AI to enhance their expertise and reach more students. Students use AI to get immediate, personalized support. Human connection between teachers and students remains central for motivation and deep understanding. This means that AI won’t replace mathematics teachers, but teachers who use AI effectively will replace those who don’t.

1.5.3 Human resource

The successful implementation of this syllabus relies heavily on qualified, competent, and motivated human resources. The following are key personnel required:

- **Teachers:** Professional and well-trained teachers with subject expertise and strong pedagogical skills in learner-cantered, participatory, and inclusive methods. Teachers should model integrity, tolerance, respect, responsibility, facilitate projects and apply effective assessment strategies.
- **Community resource persons:** Local entrepreneurs, cooperative members, financial institutions, and professionals who share real-life experiences and act as mentors during classroom sessions or field visits.
- **Parents and guardians:** They are partners in education who provide moral support, monitor learner progress, and collaborate with teachers to reinforce entrepreneurial values, creativity, responsibility, and other competences at home, ensuring holistic learner development.

- **Local Authorities and Partners:** Support schools through linkages with business development services, innovation hubs, and community-based projects that offer practical learning opportunities.

Effective collaboration among stakeholders creates an enabling learning environment that equips students with Mathematical knowledge, values, and practical competences needed for responsible citizenship, ethical behaviour, and meaningful contribution to national development.

Skills required for the Teacher of Mathematics

Effective implementation of this Syllabus requires teachers with strong subject knowledge, pedagogical skills, and interpersonal competence. As facilitators and role models, they guide students to develop Mathematical mindsets and apply competences in real-life contexts. To achieve this, teachers should demonstrate the following key skills:

a) Subject matter expertise

- Strong mastery of entrepreneurship concepts such as innovation, business opportunity identification, marketing, finance, and sustainability.
- Ability to link entrepreneurship theory with real-life examples and national development priorities, including Vision 2050 and NST2.
- Skill in relating Entrepreneurship to other disciplines such as Mathematics, ICT, Economics, and Social Studies.
- Awareness of current trends and technologies that shape the entrepreneurial landscape locally and globally.

b) Pedagogical and instructional skills

- Ability to design and implement learner-centred and participatory lessons that encourage active engagement, creativity, and critical thinking.
- Use of diverse teaching methods such as brainstorming, project-based learning, role-playing, case studies, and community engagement.

- Creativity and innovation in developing and applying diverse teaching resources, including ICT tools, realia, and community-based materials.
- Ability to adjust instruction to students' levels, ensuring inclusivity and engagement for all.

c) Assessment skills

- Proficiency in designing and applying formative and summative assessments aligned with curriculum competencies.
- Effective use of rubrics, portfolios, checklists, and peer assessments to evaluate students' competences and provide feedback that motivates students and guides improvement.
- Skill in developing valid and fair test items that assess higher-order thinking and practical skills.
- Ability to analyse assessment results and use them to guide remedial or enrichment activities.

d) Classroom management and facilitation

- Ability to establish a respectful, safe, and inclusive learning environment that promotes tolerance and values diversity.
- Proficiency in managing group work, discussions, and collaborative learning activities.
- Consistent modelling of positive behaviour, respect, punctuality, and integrity as an example for students.

e) Communication and interpersonal skills

- Strong oral and written communication to simplify complex ideas, stimulate dialogue, and foster critical reflection.
- Active listening and empathy to respond effectively to students' academic and personal needs.
- Collaboration with colleagues, parents, school leaders, and community stakeholders to support holistic learner growth.
- Ability to motivate and inspire students through encouragement and positive reinforcement.

f) Guidance and counselling skills

- Ability to provide moral, emotional, and motivational guidance to students to build confidence, resilience, and ethical conduct.
- Demonstrating passion, patience, impartiality, and genuine care in dealing with students' individual differences and challenges.
- Supporting students in exploring business ideas, career paths, and responsible life choices.

g) Professional and ethical competence

- Upholding high ethical standards, integrity, impartiality in all teaching and assessment practices, and ethical conduct, serving as a role model for students.
- Capacity to balance academic instruction with the cultivation of values and responsible citizenship.
- Demonstrating commitment to continuous professional development (CPD) through reflective practice, peer collaboration, and research.
- Modelling entrepreneurship values such as initiative, responsibility, and self-discipline.

h) Research and innovation skills

- Engagement in reflective practice and classroom-based research to improve teaching and learning.
- Adaptation of innovative approaches and new technologies to enhance learner participation.
- Connecting global entrepreneurship trends to Rwanda's socio-economic realities.
- Encouraging students to use research findings in designing projects and solving community challenges.

Teachers who possess these competences will not only deliver Mathematics effectively but will also serve as role models who inspire students to become creative, ethical, and productive contributors to Rwanda's development.

2. SYLLABUS UNITS

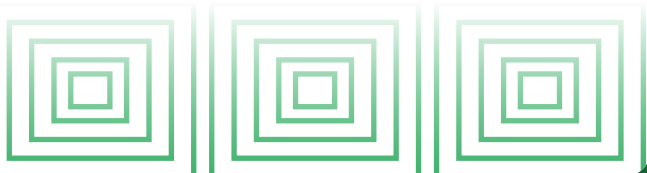
2.1 Presentation of the structure of the syllabus units

The Mathematics subject is taught and learnt in advanced level of secondary education as a core subject in S4, S5 and S6 respectively. At every grade, the syllabus is structured in topic areas, sub-topic areas where applicable and then further broken down into units to promote the uniformity, effectiveness and efficiency of teaching and learning Mathematics. The units have the following elements:

1. Each unit is aligned with the number of periods.
2. Each unit has a Key Unit Competence the achievement of which is pursued by all teaching and learning activities undertaken by both the teacher and the students.
3. Each Key Unit Competence is broken into three types of Learning Objectives as follows:
 - a) Type I: Learning Objectives relating to Knowledge and Understanding (Type I Learning Objectives are also known as Lower Order Thinking Skills or LOTS).
 - b) Type II and Type III: These Learning Objectives relate to the acquisition of skills, attitudes and values (Type II and Type III Learning Objectives are also known as Higher Order Thinking Skills or HOTS) – These Learning Objectives are actually considered to be the ones targeted by the present reviewed curriculum.
4. Each Unit has content that indicates the scope of coverage of what is to be taught and learnt in line with the stated Learning Objectives.
5. Each Unit suggests a non-exhaustive list of Learning Activities that are expected to engage students in an interactive learning process as much as possible (learner-centred and participatory approach).
6. Finally, each Unit is linked to other subjects, the Assessment Criteria and the Materials (or Resources) that are expected to be used in the teaching and learning process.

The Mathematics syllabus for Advanced level has got 7 Topic Areas: Trigonometry, Algebra, calculus, Linear algebra, Geometry, Statistics and Probability and these topic areas are found in each of the three grades of the advanced level which are S4, S5 and S6. As for units, they are 13 in S4, 10 in S5 and 9 in S6.

SENIOR FOUR



2.2 Mathematics program for secondary four

2.2.1 Key competences at the end of secondary four

After completion of secondary 4, the Mathematics syllabus will help the learner to:

- Use the trigonometric concepts and formulas in solving problems related to trigonometry
- Think critically and analyse daily life situations efficiently using mathematical logic concepts and infer conclusions
- Model and solve algebraically or graphically daily life problems using linear, quadratic equations or inequalities
- Represent graphically simple numerical functions
- Perform operations on linear transformation and solve problems involving geometric transformations
- Extend understanding, analysis and interpretation of data arising from problems and questions in daily life to include the standard deviation
- Use matrices and determinants of order 2 to solve systems of linear equations and to define transformations of 2D
- Use counting techniques and concepts of probability to determine the probability of possible outcomes of events occurring under equally likely assumptions.

2.2.2 Senior four (S4) syllabus units' content

Topic area: Algebra			Sub-topic area: Numbers and operations	
S4 Mathematics		Unit 1: Operations on the set of real numbers		Number of periods: 21
Key Unit Competence: Apply operations on real numbers, radicals, and logarithms in solving mathematical and real-life problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain absolute value and describe its properties – Recall and explain laws of powers and radicals. – State the properties of decimal logarithms – Identify relationships between exponential and logarithmic expressions – Explain how logarithmic models apply to real-life problems.	– Compute absolute values, powers, and radicals. – Simplify and rationalize expressions with radicals.	– Appreciate the importance of real numbers in daily life – Show accuracy and persistence in problem-solving.	– Introduction of the unit on the set of real numbers. – Absolute value and its properties – Powers and radicals.	– In groups, students discuss real numbers, and their operations based on the given examples. – Students use number lines to illustrate numbers individually. – Students perform exercises on simplifying powers and radicals. – Demonstrate rationalization of denominators.

	– Apply logarithmic properties to solve problems. – Model and solve real-life problems involving growth, decay, and compound interest.	– Demonstrate logical reasoning and teamwork during discussions – Value the use of mathematics in explaining real-world phenomena. – Show respect, teamwork and unity when working with colleagues.	– Operation on radicals. – Rationalization of denominators – Decimal logarithms – Properties of decimal logarithms – Solving problems about decimal logarithms – Application of decimal logarithm: Simple problems about growth, decay, magnitude of an earthquake, and compound interest.	– Solve given problems involving decimal logarithms. – In pairs, transform exponential expressions into their equivalent logarithmic forms and vice versa to understand the relationship between the two representations – Work in groups to model real-life problems involving growth, decay, and compound interest. Use GeoGebra to compare your results. – Students compute earthquake magnitude problems using logarithmic scales. Use other simulating tools (GeoGebra or PheT) to compare the your results
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Links to other subjects: **Physics** (Converting temperature from degree Celsius to degree Fahrenheit, converting seconds to minutes and vice versa); **Entrepreneurship and economics** (Organisation and computation of data from sales); **Chemistry** (The decay process); **Biology** (Growth of bacteria); **Geography** (Magnitude of an earthquake).

Assessment criteria: Students apply concepts of absolute value, powers, radicals, and decimal logarithms in analyzing, solving, and evaluating mathematical and real-life problems accurately and logically.

Materials: Graph papers, manila papers, digital technology including GeoGebra, AI, calculators

Topic area: Algebra			Sub-topic area: Mathematical logic and applications	
S4 Mathematics		UNIT 2: Binary operations		Number of periods: 14
Key Unit Competence: Apply the concepts of binary operations and algebraic structures such as groups, rings, and fields in analyzing mathematical problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify a binary operation and explain its meaning – State and describe the closure, commutative, identity, inverse, associative, and distributive properties.	– Apply the properties of binary operations to simplify and verify algebraic expressions. – Construct and use Cayley tables to test properties such as closure, commutativity, and associativity.	– Appreciate the logical structure and consistency of algebraic systems. – Develop accuracy, organization, and logical reasoning when performing mathematical operations.	– Binary operation, closure and commutative properties. – Identity and inverse properties. – Associativity and distributive properties – Cayley table of binary operation. – Algebraic structure of a Group.	– In pairs, describe a binary operation using examples on sets of numbers, identify operations that are and are not binary; – Students work in groups to test if operation is commutative, then present findings.

– Interpret a Cayley table for a given operation – Describe the basic algebraic structures and their properties. – Identify examples of Groups, Rings, and Fields within sets of numbers	– Determine whether a given set and operation form a Group, Ring, or Field. – Solve mathematical problems involving operations within algebraic structures. – Analyze and compare different algebraic structures through guided activities and exploration.	– Value mathematical precision and the role of algebraic structures in understanding real-world systems. – Show perseverance, curiosity, and teamwork in exploring abstract mathematical concepts.	– Algebraic structure of ring. – Exploring activities on the structure of Ring. – Algebraic structure of a Field. – Exploring activities on the structure of Field.	– Students identify elements for various sets through class discussion. – Students work in pairs to verify distributive, associative, and identity properties for a given operation in given sets. – Teacher facilitates in recalling the properties for a binary operation in a set to be a field. – Students test whether $\mathbb{Q}$ or $\mathbb{Q}$ with a binary operation satisfies field properties and discuss in groups why $\mathbb{Q}$ is not a field.
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Link to other subjects: Computer Science (Binary operations in programming, coding, and logic design)

Assessment criteria: Students apply properties of binary operations in analyzing and solving problems related to algebraic structures such as groups, rings, and fields accurately and logically.

Teaching and learning materials: Manila papers and markers for group work, Graphing calculator , AI and compute.

Topic area: Algebra			Sub-topic area: Mathematical logic and application	
S4 Mathematics		UNIT 3: Introduction to logic		Number of periods: 14
Key Unit Competence: Use principles of logic to analyze statements and arguments in real-life situations.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain statements, propositions, and arguments – Describe and use logical connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$).	– Construct and analyze logical arguments for validity and soundness – Build and interpret truth tables to test logical equivalence and validity of arguments	– Show accuracy, clarity, and consistency in reasoning and communication. – Demonstrate honesty and fairness in logical argumentation.	– Introduction of the unit. – Statements and propositions – Arguments. – Logical connectives (negation, conjunction, disjunction)	– Students brainstorm examples of true and false statements from everyday life and discuss why reasoning is important in problem-solving. – Teacher demonstrates how to construct a truth table for simple propositions (p, q) and for $\neg p$, $p \wedge q$, $p \vee q$ and students work in pairs to complete truth tables for given compound statements and compare results.

– Interpret truth tables for basic and compound propositions – Explain conditional and bi-conditional statements and their truth conditions. – State and apply laws of logic – Explain quantifiers (universal $\forall$ and existential $\exists$) and express their negations. – Describe propositional functions, predicates, and variables.	– Apply De Morgan's and other logical laws to simplify logical expressions. – Translate verbal statements into symbolic logic and vice versa – Use quantifiers correctly in mathematical statements and identify their negations. – Use logical reasoning to solve problems in mathematics, set theory, and real-life contexts.	– Develop curiosity and critical thinking when evaluating statements or arguments. – Appreciate the importance of logical reasoning in mathematics, science, and daily decision-making. – Work collaboratively and respect others' reasoning in discussions and group work.	– Truth tables for compound statements. – Conditional and bi-conditional statements – Tautology, contradiction and contingency – Logical equivalence and laws (De Morgan's, distributive, associative) – Quantifiers and their negations. – Introduction to propositional functions (predicate and variables)	– Construct truth tables for $\neg p$, $p \wedge q$, and $p \vee q$, compare results with classmates and explain observed patterns. – Use truth tables to predict outcomes of compound statement in group and summarize key concepts learned through the unit. – In pairs, reflect on how logic applies to mathematics and everyday decision-making. – Work in groups to solve a short logical reasoning challenge.
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– Explain applications of logic in reasoning, problem-solving, set theory, and electric circuits.	– Apply logic principles to design or analyze simple electric circuits (using AND, OR, NOT gates).		– Applications of logic to reasoning, problem-solving, set theory and electric circuits.	
Link to other subjects: Computer Science / ICT (Logic supports programming); Physics (Used to analyze electric circuits); Language: (Enhances reasoning, argument clarity, and consistency in communication); Entrepreneurship / Economics (Promotes logical decision-making and critical thinking in business analysis); Philosophy (Builds reasoning and argument evaluation skills).				
Assessment criteria: Students accurately and consistently apply logical reasoning in analyzing statements, constructing valid arguments, and solving problems.				
Teaching and learning materials: Games, puzzles, quizzes, logic gate kits, circuit boards, computers.				

Topic area: Trigonometry			Sub-topic area: Trigonometric circle and identities	
S4 Mathematics		UNIT 4: Fundamentals of trigonometry		Number of periods: 21
Key Unit Competence: Apply trigonometric circle and identities in determining trigonometric ratios and solving related problems in daily life.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain sine, cosine, tangent, and cotangent, secant, and cosecant functions. – State and use trigonometric ratios for special angles (30°, 45°, 60°, and 90°).	– Compute trigonometric ratios of given angles using a calculator or trigonometric tables. – Simplify and manipulate trigonometric expressions using standard identities.	– Show accuracy, logical reasoning, and persistence when solving trigonometric problems. – Appreciate the importance of trigonometry in solving real-life problems and in various fields such as engineering, geography, and physics.	– Introduction of sine, cosine, secant, cosecant, cotangent and tangent. – Trigonometric ratios of special angles (0°,30°, 45°, 60°, 90°) – Introduction to trigonometric identities. – Reduction to functions of positive acute angles:	– Observe a right-angled triangle and identify sides (opposite, adjacent, hypotenuse). – In groups, students derive trigonometric ratios of circle (sin, cos, tan, etc.) from side ratios. – Students complete a table showing trigonometric values of 0°, 30°, 45°, 60°, and 90° in groups and teacher guides a discussion to verify results using a calculator, or GeoGebra.

– Explain basic trigonometric identities and their use in simplifying expressions. – Describe reduction of trigonometric functions to positive acute angles (using equivalent, complementary, supplementary, and negative angles). – State and apply the sine and cosine laws for solving triangles. – Explain real-life applications of trigonometry in bearings, air navigation, angles of elevation, depression, and inclined planes.	– Solve triangles using sine and cosine laws. – Apply trigonometric concepts to solve practical problems in navigation, bearings, and height-distance measurements. – Draw and interpret diagrams involving angles of elevation, depression, and inclinations.	– Develop curiosity and confidence in exploring geometric and spatial relationships. – Work collaboratively and responsibly while performing practical trigonometric tasks. – Demonstrate honesty and precision in recording and interpreting measurements.	– Equivalent angles – Negative/ Opposite angles. – Complementary angles. – Supplementary angles. – Solving a triangle by cosine and sine law. – Applications of trigonometry: – Bearing – Air navigation – Angle of elevation – Angle of depression – Inclined plane.	– Simplify expressions using identities in pairs or in small groups. – Investigate how trigonometric functions behave for complementary, supplementary, and negative angles using the unit circle. – In groups, use quadrant signs to find equivalent acute angles. – Practice exercises on converting angles to the equivalent acute angles and simplifying. – Students use maps or compasses to find directions and angles between points. – Students discuss how trigonometry supports architecture, surveying, and aviation.
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Link to other subjects: **Physics** (Analyzing forces, motion on inclined planes, waves, vectors, and light reflection/refraction problems); **Geography** (Map reading, navigation, and determining distances or heights of geographical features using angles of elevation and depression); **Construction** (Design and interpret plans, roofs, bridges, and buildings with correct angles and slopes); **Aviation and Navigation** (Determining bearings, distances, and directions for pilots and navigators).

Assessment criteria: Students apply trigonometric ratios, identities, and laws in analyzing and solving mathematical and real-life problems involving triangles, bearings, navigation, angles of elevation and depression, and inclined planes accurately and logically.

Teaching and learning materials: Rulers, protractors, set squares, magnetic compasses, measuring tapes, and clinometers for angle and height measurements, calculators, mathematics software (e.g., GeoGebra, Desmos and AI).

Topic area: Algebra			Sub-topic area: Equations and inequalities	
S4 Mathematics		UNIT 5: Linear and quadratic equations and inequalities		Number of periods: 21
Key Unit Competence: Solve daily life problems algebraically or graphically using linear and quadratic equations or inequalities.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain and interpret inequalities involving absolute values. – Explain the concept of parametric equations and inequalities in one unknown.	– Solve inequalities involving absolute values using graphical and algebraic methods – Express and manipulate parametric equations to eliminate parameters and find relationships between variables	– Show accuracy, persistence, and logical reasoning in solving mathematical problems. – Develop confidence and independence in exploring different solution methods.	– Parametric equations in one unknown – Parametric inequalities in one unknown – Solving equations reducible to quadratic form: bi-quadratic equations	– Students are guided on how to eliminate parameters in parametric equation of a line to find the Cartesian equation; the, to use dynamic geometry software (e.g., GeoGebra) to visualize how parameters affect curves. – Students solve and compare inequalities for specific parameter values.

– Recognize and classify equations reducible to quadratic form (bi-quadratic, nested, irrational, and reciprocal equations). – State and apply methods for solving quadratic inequalities and simultaneous quadratic equations in two unknowns. – Clarify the relationship between equations, inequalities, and their graphs.	– Transform complex expressions into quadratic form and solve them accurately. – Solve bi-quadratic, nested, irrational, and reciprocal equations step-by-step – Determine and interpret solutions to quadratic inequalities using number lines or graphs – Solve simultaneous quadratic equations both algebraically and graphically	– Appreciate the practical importance of equations and inequalities in real-life decision-making. – Demonstrate collaboration and effective communication when working in groups. – Display intellectual curiosity and creativity in problem-solving.	– Solving equations reducible to quadratic form: Nested equations, irrational equations, reciprocal equations – Quadratic inequalities – Parametric equations/ inequalities in one unknown – Inequalities involving absolute values.	– In groups, explore the guided examples on solving bi-quadratic (x^4), irrational (square roots), nested, and reciprocal forms. Practice exercises with peer correction and teacher feedback. – Students review the definition and properties of absolute value. Students solve simple absolute value equations and inequalities on the board, they solve and present solutions on a number line and create real-life examples. – Students solve simultaneous quadratic equations using substitution and elimination, they practice algebraic methods in group.
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– Describe how equations and inequalities model real-life problems in various contexts.	– Use equations and inequalities to solve real-life problem situations (e.g., projectile motion, optimization, and economic modeling). – Use ICT tools or graphing software to visualize and verify results.	– Uphold honesty and precision in presenting mathematical results.	– Simultaneous quadratic equations in two unknowns – Solving problems from real life situations involving equations and inequalities.	– Students model problems mathematically and solve using appropriate methods and then do group presentations of solutions and reasoning.
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Link to other subjects: **Physic** (Model motion such as displacement, velocity, acceleration, energy relationships, and projectile motion), **Economics / Entrepreneurship** (Express cost, revenue, and profit relationships, and to analyze inequalities in resource allocation, market trends, and optimization); **Geography** (Applied in modeling population growth, pollution spread, and environmental data); **Chemistry** (Expressing concentration, reaction rates, and equilibrium conditions that involve relationships between variables); **Biology** (to model growth rates of populations or bacteria).

Assessment criteria: Students apply concepts of equations and inequalities in modeling and in solving mathematical and real-life problems accurately and logically.

Teaching and learning materials: Graph paper, rulers, protractors, Scientific calculators, computers with graphing software (e.g., GeoGebra, Desmos and AI).

Topic area: Analysis			Sub-topic area: Functions	
S4 Mathematics		UNIT 6: Polynomial, rational and irrational functions		Number of periods: 21
Key Unit Competence: Analyze polynomial, rational, and irrational functions by determining their domains, representing them graphically, and solving related problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the concept of a polynomial and factorization of polynomials. – State and interpret the domain and range of a function. – Identify the domain of definition for polynomial, rational, and irrational functions.	– Factorize polynomials accurately using different methods (grouping, formulas, etc.). – Determine domain and range of given functions both algebraically and graphically. – Perform operations on functions and construct composite and inverse functions.	– Show logical reasoning, accuracy, and persistence when solving algebraic problems – Develop appreciation for the use of functions and polynomials in modeling real situations	– Recall on factorization of polynomials. – Definition of domain and range of a function – Determination of domain of definition for polynomial, and rational and irrational functions – Operations on functions	– Recall previous methods of factorization and work in pairs to factorize different polynomials using various techniques. – Students discuss how factorization helps in simplifying rational functions and Teacher guides them to solve step-by-step examples.

– Describe operations on functions – Explain composite and inverse functions and determine the parity (evenness/oddness) of a function. – Explain the process of decomposing rational functions into partial fractions for different types of denominators.	– Test and classify functions as even, odd, or neither. – Decompose rational expressions into partial fractions involving linear and quadratic factors. – Apply polynomial, rational, and irrational functions to solve real-life problems in various fields.	– Demonstrate curiosity and creativity in manipulating algebraic expressions – Work collaboratively and share strategies respectfully during group tasks – Exhibit confidence, honesty, and responsibility when handling mathematical operations and solutions.	– Composite and inverse functions – Parity of a function – Partial fractions decomposition for denominator containing different linear factors – Partial fractions decomposition for denominator containing repeated linear factors – Partial fractions decomposition for denominator containing linear factor and repeated linear factors	– In groups, students discuss how each operation affects the domain and compare results obtained algebraically and graphically. – Students observe symmetry on graphs to determine if functions are even or odd and discuss real-life examples of even and odd functions. – Students engage in teacher-guided examples to decompose rational functions into partial fractions and compare their results with those generated by an AI.
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– Recognize the importance and applications of rational and irrational functions in real-life contexts.			– Partial fraction decomposition for denominator containing linear and quadratic (which cannot be factored) factors – Applications of rational and irrational functions in diary life.	– Through teacher guide, students identify real-life problems modeled by rational or irrational functions; Students formulate and solve equations representing those problems, present and discuss findings in groups.
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Links to other subjects: **Physics** (Modeling the fall of a ball); **Economics** (To represent the cost of producing “x” units of a commodity, or marginal cost); **Chemistry** (Use polynomial to express the rate of reaction in chemistry).

Assessment criteria: Students apply concepts of polynomials, rational, and irrational functions in analyzing, performing operations, and solving mathematical and real-life problems accurately and logically.

Teaching and learning materials: Graph paper, rulers, protractors, scientific calculators, computers with mathematical software (e.g., GeoGebra, Desmos, AI tool or graphing calculators).

Topic area: Algebra			Sub-topic area: Numbers and operations	
S4 Mathematics		UNIT 7: Limits of polynomial, rational and irrational functions		Number of periods: 14
Key Unit Competence: Apply correctly limits of polynomial, rational and irrational functions in solving related problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the neighborhood of a real number and the concept of a limit of a variable – State and interpret the limit of a function both algebraically and graphically	– Evaluate limits of functions algebraically, graphically, and numerically – Determine left-hand and right-hand (one-sided) limits of given functions – Compute infinite limits and limits at infinity using appropriate techniques	– Show persistence, accuracy, and logical reasoning when solving limit problems – Demonstrate curiosity in exploring graphical behavior of functions near critical points	– Neighborhood of a real number and limit of a variable – Definition and graphical interpretation of limit of a function – One-sided limits – Infinite limits – Limits at infinity	– Students are guided to recall prior knowledge of real numbers and intervals. – Students explore examples in which the value of a variable approaches a specific number, using a number line or graph in GeoGebra to visualize the neighborhood around a point. Working in pairs, they describe and discuss what happens as the variable approaches the given number.

– Distinguish between one-sided limits, infinite limits, and limits at infinity – Explain the Squeeze theorem and its use in finding limits. – Identify and describe indeterminate cases and how they occur in limit evaluation – Explain the applications of limits in determining continuity, discontinuity, and asymptotes of functions.	– Apply the Squeeze theorem to find limits of complex functions. – Identify and resolve indeterminate cases using algebraic manipulation – Use limits to analyze the continuity and discontinuity of functions – Apply limits to determine vertical, horizontal, and oblique asymptotes of rational and irrational functions.	– Appreciate the role of limits as a foundation for calculus and real-world modeling – Work collaboratively and share ideas respectfully. – Exhibit confidence and responsibility in interpreting mathematical results.	– The Squeeze theorem and Operations on limits – Indeterminate cases – Applications of limits on continuity and discontinuity of a function – Applications of limits on asymptotes.	– Students use calculators, GeoGebra to estimate limits graphically then discuss and formulate the formal definition of a limit. – Teacher guides students to summarize the concept of limits and their uses in analyzing functions. – Students work in groups to solve real life problems on limits and continuity. – Students present group findings and reflect on challenges faced during learning.
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Link to other subjects: **Physics** (Studying motion, speed, and acceleration); **Chemistry** (Reaction rates and concentration changes over time); **Biology** (Population growth models, enzyme reaction rates, and logistic growth); **Economics** (Limits describe marginal cost, marginal revenue, and continuous change in supply and demand); **Geography** (Interpreting graphs of climate change, elevation curves, or population density).

Assessment criteria: Students apply the concepts and techniques of limits in analyzing, interpreting, and solving real life problems involving continuity, asymptotes, and the behavior of functions accurately.

Teaching and learning materials: Graph papers, rulers, coordinate grids, digital technologies including GeoGebra, AI and scientific calculators.

Topic area: Analysis			Sub-topic area: Limits, differentiation and integration	
S4 Mathematics		UNIT 8: Differentiation of polynomials, rational, irrational functions and their applications.		Number of periods: 35
Key Unit Competence: Apply derivatives of polynomial, rational, and irrational functions in solving related mathematical and real-life problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe the derivative of a function and explain the concept of differentiation from first principles – State and explain the rules of differentiation	– Differentiate functions using first principles and standard rules – Compute higher-order derivatives of given functions – Determine equations of tangents and normal to curves	– Show curiosity and critical thinking when exploring how functions change	– Concepts of derivative of a function – Rules of differentiation: Sum or difference, The product, the quotient and the chain rule. – High order derivatives – Applications of differentiation:	– In groups, students compute derivatives from first principle for simple functions; Students discuss to derive the general formula for derivatives.

– Describe higher-order derivatives and their meanings – Explain geometric interpretations of derivatives (tangent, normal) – State and interpret Lagrange's and Rolle's theorems – Explain L'Hôpital's rule and its application in evaluating limits	– Apply Lagrange's and Rolle's theorems to solve mathematical problems – Use L'Hôpital's rule to evaluate indeterminate limits. – Analyze functions to find stationary points, concavity, and inflection points	– Demonstrate accuracy, patience, and logical reasoning when performing differentiation. – Appreciate the importance of derivatives in modeling and solving real-life problems (e.g., motion, economics, and engineering) – Develop perseverance and confidence in tackling challenging mathematical problems	– Geometric interpretation of derivatives: Equation of the tangent to a curve. – Geometric interpretation of derivatives: Equation of normal to a curve. – Mean value theorem for derivatives: Lagrange's theorem – Rolle's theorem – Hospital's theorem – Variations of a function and critical points: stationary and turning point – Concavity of a function and inflexion points – Table of variation of a function	– The teacher guides students to demonstrate the rules of differentiation using examples on the board, after which students practice applying each rule to various types of functions. In groups, students identify the appropriate rule to apply in mixed expressions, followed by peer correction, presentation of their findings, and feedback. – The teacher guides students in finding the equations of the tangent and normal at a given point of a curve. Students then plot the curve and draw the tangent and normal lines either manually or by using graphing software such as GeoGebra.
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	– Construct and interpret a table of variation. – Sketch polynomial, rational, and irrational functions using derivative properties – Solve rate of change and optimization problems using differentiation techniques.	– Value the use of mathematical reasoning and graphical representation in problem-solving.	– Curve sketching of a polynomial function – Curve sketching of a rational function – Curve sketching of an irrational function – Applications of differentiation: – Gradient as a measure of rate of change – Kinematic meaning of derivatives – Optimization problems.	– Teacher presents the theorems with visual illustrations on graphs then students discuss the conditions required for each theorem and students solve problems applying the theorems in group. – Teacher provides contextual problems (e.g., maximizing area, minimizing cost, rate of change in motion). – Students solve real-life optimization and rate problems individually or in groups. – Students present and justify their solutions. – Class discussion on how derivatives model real-world changes.
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Link to other subjects: **Physics:** (Velocity, acceleration, kinematics, dynamics, and mechanics to model real phenomena); **Economics:** (Finding marginal cost, marginal revenue, and profit maximization problems); **Biology** (Modeling population growth rates, rates of chemical reactions in cells, and spread of diseases over time); **Chemistry** (Representing rates of reaction and concentration changes with respect to time); **Geography** (Studying rate of population growth, changes in temperature, and gradient of slopes on geographical surfaces); **Entrepreneurship** (Minimizing production costs and maximizing profits).

Assessment criteria: Students apply differentiation concepts in analyzing, interpreting, and solving mathematical and real-life problems involving rates of change, curve behavior, optimization, and graphical representation accurately and logically.

Teaching and learning materials: Manila paper, graph paper, graphs showing tangents, normal, concavity, and inflection points, and digital technologies including calculators, GeoGebra and AI.

Topic area: Linear Algebra			Sub-topic area: Vectors	
S4 Mathematics		UNIT 9: Vectors space of real numbers.		Number of periods: 14
Key Unit Competence: Determine the magnitude of vectors, angle between two vectors and the dot product of two vectors.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe vectors and vector spaces – Describe operations on vectors (addition, scalar multiplication) – Explain the properties of vector operations in vector spaces – Define sub-vector spaces, linear combination, basis, and dimension.	– Perform vector addition, scalar multiplication, and other vector operations accurately – Identify and verify sub-vector spaces – Express vectors as linear combinations of given vectors. – Determine basis and dimension of vector spaces	– Show curiosity and logical reasoning in exploring vector spaces. – Demonstrate precision, accuracy, and attention to detail in vector calculations.	– Definitions and operations on vectors in the Vector spaces R^2 – Properties of operations on vectors in Vector spaces R^2 – Sub-vector spaces of the Vector spaces R^2	– Teacher guides students to use real life examples and introduce vectors (e.g., forces, velocity) and they discuss vector representation, notation and perform vector addition, subtraction, and scalar multiplication.

– Describe Euclidean vector space and its features – Explain the dot product, its properties, magnitude of vectors, and the angle between two vectors.	– Compute dot product, magnitude, and angles between vectors – Solve problems in Euclidean vector space efficiently.	– Appreciate the relevance of vectors in mathematics, physics, and engineering. – Develop persistence and confidence in solving vector-related problems. – Work collaboratively in performing exercises involving vector operations.	– Linear combination of vectors the Vector spaces R^2 – Basis and dimension of a vector space – Introduction to Euclidian Vector space R^2 – Dot product and properties – Modulus or magnitude of vectors – Angle between two vectors	– In groups, discuss and explore properties of a vector space (closure, commutativity, associativity, and distributivity) through examples. – In pairs, students discuss necessary conditions for subspaces. – Teacher are guided to discuss the meaning of Euclidean vector space and make geometric interpretation; they plot vectors on a coordinate system and observe geometric relationships between them. – Individually or in pairs, students calculate dot products and then discussions to verify properties and applications.
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				– Students discuss the meaning of magnitude of vectors and compute magnitude of various vectors.
Link to other subjects: Physics (Representing forces, velocity, acceleration, and displacement); Engineering (Analyzing forces, designing structures, and representing directions); Geography (Representing wind velocity, ocean currents, and movement directions on maps).				
Assessment criteria: Students accurately and logically apply concepts and operations of vectors in analyzing, computing, and solving mathematical and real-life problems involving vector spaces, linear combinations, dot products, magnitudes, and angles between vectors.				
Teaching and learning materials: Rulers, compasses, and protractors for geometric representation of vector, mathematics software like GeoGebra, Desmos AI, or MATLAB.				

Topic area: Linear Algebra			Sub-topic area: Linear transformations in 2D	
S4 Mathematics		UNIT 10: Matrices and determinants of order 2		Number of periods: 14
Key Unit Competence: Use matrices and determinants of order 2 in solving systems of linear equations				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe matrices of linear and geometric transformation – Explain operations on matrices: equality, addition, scalar multiplication, multiplication, transpose, and inverse	– Compute and represent linear and geometric transformations using matrices. – Perform matrix operations accurately and efficiently. – Calculate determinants and use them to solve problems	– Appreciate the importance of matrices in modeling and solving problems. – Demonstrate curiosity and analytical thinking when exploring applications of matrices	– Linear transformation in two dimensions – Matrix of a linear transformation – Operations on matrices and their properties: Equality and Addition of matrices.	– Explore examples of matrices for linear and geometric transformations. – In pairs, perform operations of matrices: equality, addition, scalar multiplication, and multiplication of matrices using examples. – Teacher guides students to perform exercises on matrix operations individually and in groups.

– Describe determinants of 2×2 matrices and their properties – Identify applications of determinants in solving linear systems and transformation problems.	– Apply matrix operations and determinants to real-life and mathematical problems.	– Develop confidence and responsibility when handling matrix-related tasks.	– Operations on matrices and their properties: Multiplication of matrices. – Determinant of a matrix of order 2: Definition and properties. – Operations on matrices and their properties: Transpose and inverse of a square matrix. – Applications of determinants.	– Explore the transpose and inverse of a square matrix and show step-by-step calculations. – Teacher guides students to explain the determinant of a 2×2 matrix and explore its properties through examples. – Students apply determinants to solve linear systems and transformation problems. – Use visual aids, software such as GeoGebra, and AI to demonstrate matrices in geometric transformations and to perform various matrix operations. – Workout tasks that combine multiple matrix operations to model real-life situations then, discuss and explain their solutions.
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Link to other subjects: **Physics** (Representing rotations, reflections, and transformations in mechanics and optics); **Statistics** (In data representation, regression analysis).

Assessment criteria: Students apply concepts of matrices and determinants in analyzing, computing, and solving mathematical and real-life problems involving linear and geometric transformations.

Teaching and learning materials: Computers or tablets with GeoGebra or other matrix software, calculators, ruler, compass, protractor, graph paper and AI.

Topic area: Linear Algebra			Sub-topic area: Linear transformations in 2D	
S4 Mathematics		UNIT 11: Operations on linear transformation in 2D.		Number of periods: 14
Key Unit Competence: Apply linear and geometric transformations in two dimensions in solving mathematical and real-life problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the concepts of linear and geometric transformations in two dimensions. – Describe the properties of orthogonal and orthonormal transformations	– Apply linear and geometric transformations to solve mathematical problems – Construct and interpret reflections, rotations, and symmetries on coordinate planes – Determine the kernel and range of given linear transformations.	– Appreciate the importance of transformations in solving geometric and real-life problems – Show accuracy, patience, and critical thinking when performing transformations	– Geometric transformation in 2D: Reflection. – Central symmetry. – Rotation around the origin (0,0) of axes. – Orthogonal transformations. – Operations on transformations. – Matrix of a composite linear transformation in 2D.	– Teacher introduces the concept of transformations through visual examples and real-life illustrations such as mirror images or object rotations. – Students explore linear and geometric transformations using coordinate grids or digital tools like GeoGebra.

– Identify the types of symmetry in geometric figures – Identify the identical (identity) transformation and its effect on figures.	– Represent transformations algebraically and geometrically.	– Demonstrate logical reasoning and consistency in problem-solving.	– Kernel of a linear transformation. – Range of a linear transformation.	– Students are guided to demonstrate reflections about the x-axis, y-axis, and central symmetry through step-by-step examples. – Teacher engages students in hands-on activities to perform rotations and identify identical (identity) transformations. – Teacher facilitates small group discussions where students investigate the properties of orthogonal and orthonormal transformations. – Teacher provides exercises for students to calculate and interpret the kernel and range of linear transformations.
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				– Teacher assigns problem-solving tasks that combine multiple transformations to find resulting images. – Teacher guides students to connect transformation concepts to real-world applications such as in graphic design and physics.
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Link to other subjects: **Physics** (Explaining motion, forces, and optics); **Computer science** (In computer graphics, image processing, animations, and programming of visual simulations); **Geography** (Map reading and coordinate transformations relate to geometric transformations used in scaling and projection); **Art and design** (In patterns, design, and architectural drawings).

Assessment criteria: Students apply concepts of linear and geometric transformations in 2D in analyzing, computing, and solving mathematical and real-life problems involving reflections, rotations, symmetries, orthogonal transformations, and the kernel and range of transformations accurately.

Teaching and learning materials: Mathematical instruments (compasses, protractors, ruler), graph paper, coordinate grids and marker, dynamic geometry software (e.g., GeoGebra) and AI.

Topic area: Statistics and probability			Sub-topic area: Descriptive statistics	
S4 Mathematics		UNIT 12: Measures of dispersion		Number of periods: 7
Key Unit Competence: Analyze measures of dispersion for data on real life situations in informing a conclusion.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Recall and explain measures of central tendency and box plot representation. – Describe variance, standard deviation, and coefficient of variation.	– Construct and interpret box plots using given data – Calculate variance, standard deviation, and coefficient of variation accurately.	– Appreciate the importance of summarizing data for analysis and decision-making – Demonstrate accuracy, patience, and attention to detail statistical calculations	– Recall on measures of central tendency – Quartiles. – Presentation of quartiles on box plots. – Variance – Standard deviation (including combined set of data)	– In groups, review of measures of central tendency (mean, median, and mode) and their importance in data analysis. – Demonstrate how to construct and interpret box plots using example data sets. – Calculate the interquartile range and use it to construct box plots individually and in groups. – Explore variance, standard deviation, and coefficient of variation with step-by-step examples. – In pairs, compute variance and standard deviation for individual and combined data sets.

– Explain the application of measures of dispersion in data interpretation – Identify relationships between measures of central tendency and measures of dispersion.	– Apply measures of dispersion to analyze and interpret data sets meaningfully. – Compare different data sets using appropriate statistical measures.	– Show critical thinking and curiosity in exploring patterns and trends in data – Develop responsibility and logical reasoning in handling and interpreting data.	– Coefficient of variation – Application of measures of dispersion in data interpretation.	– Teacher facilitates group discussions on the interpretation of measures of dispersion in real-life contexts. – Perform problem-solving activities to use measures of dispersion to compare and analyze multiple data sets. Use charts, graphs, and other visual aids to illustrate the relationship between measures of central tendency and dispersion, supported by digital tools such as Excel, or GeoGebra. Present and explain their findings, highlighting trends, patterns, and anomalies. – Teacher guides students to make reflective discussions on the significance of measures of dispersion in making informed decisions.
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Link to other subjects: **Economics** (Analyzing economic data, market trends, and risk assessment); **Biology** (Analyzing population data, experimental results, and biodiversity studies); **Physics** (In measurement errors, experimental data analysis, and uncertainty estimation); **Geography** (Analyzing spatial data, climate variability, and demographic distribution); **Social studies** (Interpreting survey data, census results, and social research findings).

Assessment criteria: *Students critically and logically select and apply appropriate counting techniques, as well as measures of central tendency and dispersion, to model, solve, and justify solutions to mathematical problems and authentic real-world situations.*

Teaching and learning materials: *Real-life data sources (e.g., survey results, experimental data), computers or tablets with spreadsheet software (e.g., Excel, Google sheet, AI), and calculators.*

Topic area: Statistics and probability			Sub-topic area: Combinatorial analysis and probability	
S4 Mathematics		UNIT 13: Combinatorics		Number of periods: 21
Key Unit Competence: Use combinations and permutations in determining the number of ways a random experiment occurs.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the basic counting principles (addition and multiplication). – Describe Venn diagrams, tree diagrams, and contingency tables and their uses in organizing data.	– Construct and interpret Venn diagrams, tree diagrams, and contingency tables to solve counting problems. – Apply the multiplication principle to determine the number of possible outcomes in different situations.	– Appreciate the importance of systematic and logical thinking in problem solving. – Develop accuracy and persistence when solving counting and combinatorial problems.	– Introduction to counting techniques – Venn diagram – Tree diagrams – Contingency table – Multiplication principles – Arrangements without repetitions – Arrangements with repetitions or conditions	– Brainstorm and share real-life situations where you need to count the number of possible outcomes. – Work in groups to discuss and review what you already know about sets and basic operations, connecting these ideas to counting techniques. – Practice drawing and interpreting Venn diagrams, tree diagrams, and contingency tables to represent and solve counting problems.

– State and distinguish between arrangements and combinations (with or without repetitions or conditions). – Explain the concept of circular permutation. – Describe Pascal's triangle and its relationship with binomial expansion. – State and apply the Binomial Theorem.	– Calculate permutations and combinations with or without repetitions or conditions. – Use Pascal's triangle to determine coefficients in binomial expansion. – Solve real-life problems involving binomial expansion and counting techniques.	– Show interest and curiosity in exploring different counting strategies. – Value teamwork and discussion in discovering various ways to approach combinatorial problems.	– Permutations with or without repetitions – Circular permutation – Combinations: Definitions and properties – Combinations with repetitions or conditions – Pascal's triangles – Binomial expansion – Guided practice on binomial expansion	– Participate in hands-on activities, such as arranging different colored objects or letters, to explore and discover the principles of arrangements and permutations. – Physically arrange yourselves sitting around a table to understand how circular permutations work. – Work in small groups to compute combinations using formulas and compare your results with real-life examples. – Construct Pascal's triangle together and work to identify patterns within the coefficients. – Take part in problem-solving sessions to expand binomials, using both Pascal's triangle and the Binomial Theorem. – Complete guided practice exercises on each topic, working individually or in groups, and then discuss the solutions as a class.
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				– Undertake enrichment tasks, such as designing a mini-project or creating a puzzle that involves counting principles or binomial expansion.
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Link to other subjects: **Computer Science:** (in algorithm design, data arrangement, and calculating possible combinations in coding, password generation), **Physics:** (in calculating possible arrangements of particles), **Geography:** (in statistical mapping, survey sampling, and analyzing combinations of factors affecting natural phenomena), **Entrepreneurship:** (in planning schedules, organizing teams, product packaging combinations, and optimizing available resources).

Assessment criteria: Students correctly apply counting principles, permutations, combinations, and binomial expansion to solve mathematical and real-life problems with accuracy and logical reasoning.

Teaching and learning materials: Manila papers, graph paper, ruler, digital components including calculators and AI.

SENIOR FIVE

2.3 Mathematics program for secondary five

2.3.1 Key competencies at the end of secondary five

After completion of secondary 5, the mathematics syllabus will help the learner to:

- Study and represent graphically a numerical function.
- Extend the use of the trigonometric concepts and transformation formulas to solve problems involving trigonometric equations, inequalities and/or trigonometric identities.
- Use arithmetic, geometric and harmonic sequences and their convergence, to understand and solve problems arising in various contexts.
- Apply the meaning of coordinates in solving problems involving points, lines and shapes in 2D space.
- Solve equations involving logarithms or exponentials and apply them to model and solve related problems.
- Apply theorems of limits and formulas to solve problems involving differentiation including optimisation.
- Study linear dependence of vectors of $\mathbb{R}^3$ and perform operations on linear transformations of $\mathbb{R}^3$ using vectors.
- Extend the use of matrices and determinants to order 3 to solve problems in various contexts.
- Use algebraic representations of lines, spheres and planes in 3D space and solve related problems.
- Extend the understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.
- Solve problems using Bayes theorem and data to make decisions about likelihood and risk.
- Determine algebraic representations of lines, straight lines and circles in 2D

2.3.2 Mathematics units for secondary five

Topic area: Trigonometry			Sub-topic area: Trigonometric circle and identities	
S5 Mathematics		UNIT 1: Trigonometric Formulae, Equations and Inequalities		Number of periods: 28
Key Unit Competence: Use trigonometric formulae, equations, and inequalities in solving mathematical and real-life problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Show how to use transformation formulas to simplify the trigonometric expressions. – Extend the concepts of trigonometric ratios and their properties to trigonometric equations.	– Apply the transformation formulas to simple trigonometric expressions. – Use trigonometric functions and equations to model and solve problems involving trigonometry concepts.	– Develop precision and logical reasoning when manipulating trigonometric expressions.	– Period of trigonometric functions. – Transformation formulas: addition and subtraction formulas. Double-angle and half-angle formulas. – Sum, difference and product formulas.	– Recall basic trigonometric functions and determine the periods of each. – Demonstrate addition and subtraction formulas using step-by-step examples; students practice in pairs.

– Analyze and discuss the solution of trigonometric inequalities. – Explain the structure and solution methods for trigonometric equations and inequalities. – Describe real-life applications of trigonometric formulas, including integration and harmonic motion in physics	– Apply trigonometric functions and equations to perform so far operations in calculation, in integration and solving problems including harmonic motion in physics. – Solve any other problems related to trigonometry.	– Appreciate the relevance of trigonometry in real-life situations, especially in physics. – Show persistence and curiosity when exploring multiple solution methods. – Appreciate the relationship between trigonometry and other subjects.	– Trigonometric equations: Trigonometric equation reducible to the form: $\sin(x + \alpha) = k$, $\cos(x + \alpha) = k$, and $\tan(x + \alpha) = b$ for $k \leq 1$ and a real number ($b \in \mathbb{R}$). – Trigonometric equation reducible to the form $\sin nx = k$. – Trigonometric equation of the form $a \sin x + b \cos x = c$. – Trigonometric inequalities. – Applications: Simple harmonic motion in physics, Refraction of light.	– In groups, apply double-angle and half-angle formulas to simplify expressions and solve related problems. – Apply basic rules to demonstrate the sum, difference, and product formulas, and complete guided exercises to reinforce understanding. – Present trigonometric equations and solve progressively challenging examples individually or in small groups.
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				– Explore trigonometric inequalities, including graphical representation on the unit circle and solving algebraically. – Provide, model and solve application problems on harmonic motion in physics. – Conclude with a reflection discussion on the relevance of trigonometry in science and engineering contexts.
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Link to other subjects: **Physics:** (In harmonic motion, analyzing waves, oscillations, forces, and vector components), **Geography:** (In surveying, mapping, and navigation, especially using angles, bearings, and distance calculations), **Computer Science** (In graphics, simulations, and animations, especially for periodic movements or rotations), **Construction** (In calculating forces on inclined planes, roof angles, and ramps).

Assessment criteria: Students solve trigonometric problems using transformation formulas, equations, and inequalities in modeling real-life situations in physics, construction, and engineering accurately.

Teaching and learning materials: Geometric instruments (ruler, T-square, compass), graph paper, calculators or software (e.g., Desmos, GeoGebra) or AI for visualizing equations and inequalities, pendulums, springs.

Topic area: Algebra			Sub-topic area: Number patterns	
S5 Mathematics		UNIT 2: Sequences		Number of periods: 21
Key Unit Competence: Use arithmetic, geometric, and harmonic sequences, including their convergence in analyzing mathematical and real-life problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain sequence and distinguish between convergent and divergent sequences. – Explain monotone sequences and their characteristics. – State the definitions of arithmetic, geometric, and harmonic sequences.	– Determine the general term of arithmetic and geometric sequences. – Calculate arithmetic, geometric, and harmonic means. – Compute the sum of the first “n” terms of arithmetic and geometric sequences.	– Develop accuracy, logical reasoning, and persistence in working with sequences. – Appreciate the importance of sequences in modeling real-life situations.	– Meaning of sequences. – Monotone sequences – Arithmetic sequences: (definition, common difference) – General term of an arithmetic sequence – Arithmetic means of a sequence	– Demonstrate arithmetic sequences, including general term, common difference, arithmetic means, and sum of “n” terms. Students practice examples individually and in pairs.

– Explain the concepts of general term, arithmetic/geometric means, and sum of “n” terms. – Understand applications of sequences in real-life problems.	– Analyze sequences to identify convergence, divergence, and monotonicity. – Apply sequences to solve practical, real-life problems.	– Show curiosity and willingness to explore patterns and relationships in numerical data.	– Sum of n first terms of an arithmetic sequence – Harmonic sequence. – Geometric sequences: definition, common difference), general term of a geometric sequence – Geometric means of numbers. – Sum of n first terms of a geometric sequence – Convergent and divergent sequences. – Application of sequences in solving real life problems.	– Introduce geometric sequences, showing general term, geometric means, and sum of “n” terms. Students solve guided exercises. – Explore harmonic sequences and relate them to real-life contexts such as speeds or rates. – Work in groups to analyze monotone sequences and determine their convergence or divergence, then compare results with those generated by an AI.
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				– Provide real-life application problems, such as calculating savings over time, population models, or interest accumulation, and guide students to solve them using sequences.
Link to other subjects: Physics (in modeling harmonic motion, wave patterns, or periodic phenomena), Biology (in modeling population growth, decay, or cyclic biological events), Chemistry (in reaction rates, periodic trends, or concentrations over time). Finance / Economics (in calculating savings, loans, investments, and compound interest).				
Assessment criteria: Students identify different types of sequences, find their general terms and sums, and use these skills to solve real-life problems involving patterns, growth, and decay.				
Teaching and learning materials: Counters or objects for hands-on demonstration of sequence patterns, Graphing software (e.g., GeoGebra, Desmos) or AI.				

Topic area: Algebra			Sub-topic area: Equations and inequalities	
S5 Mathematics		UNIT 3: Logarithmic and Exponential Equations		Number of periods: 14
Key Unit Competence: Apply exponential or logarithmic functions and equations in solving mathematics and real-life problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain exponential and logarithmic functions and identify their properties. – Explain the relationship between exponential and logarithmic functions. – State and apply the laws of exponents and logarithms. – Describe the nature of exponential growth and decay processes.	– Construct and interpret graphs of exponential and logarithmic functions. – Simplify and manipulate exponential and logarithmic expressions. – Solve exponential and logarithmic equations analytically and graphically.	– Appreciate the usefulness of exponential and logarithmic models in real-life problem-solving. – Show persistence and logical reasoning when solving equations and modeling data.	– Exponential functions – Logarithmic functions – Exponential equations – Logarithmic equations – Applications of exponential and logarithmic equations:	– Solve exponential equations using properties of exponents and logarithms, refer to step-by-step problem-solving methods.

– Recognize real-life situations that can be modeled by exponential and logarithmic equations.	– Apply exponential and logarithmic models to real-life problems such as interest, population growth, radioactive decay, mortgages, earthquakes, carbon dating, and alcohol-risk analysis. – Interpret solutions in context and draw meaningful conclusions	– Develop accuracy and critical thinking in mathematical computations. – Demonstrate teamwork, communication, and a positive attitude toward applying mathematics in daily life.	– Interest and Population growth problems. – Radioactive decay problems – Mortgage problems – Earthquake problems and Carbon dating problems – Problems about alcohol and risk of car accident.	– Work individually and in pairs to solve exponential equations, verify results graphically, and explain the reasoning behind their solutions. – The teacher presents real-life contexts for applying exponential and logarithmic equations and then provide a project to explore more applications such as compound interest, population growth, decay, mortgages, earthquakes, and alcohol risk.
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				– Work in groups to analyze mathematical problems, choose the right model (exponential or logarithmic), perform calculations, draw graphs, and present results. Discuss interpretations and implications of results in real life.
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Link to other subjects: **Physics** (Radioactive decay, half-life, earthquake magnitude); **Biology** (Population growth and decay), **Economics & Entrepreneurship**(Compound interest, investments, and mortgage problems); **Chemistry**(Reaction rate modeling and carbon dating); **ICT** (Use of graphing calculators or software for data modeling).

Assessment criteria: Students apply concepts of exponential and logarithmic functions to analyze, compute, and solve mathematical and real-life problems involving growth and decay, interest, population, radioactive decay, earthquakes, carbon dating, and risk assessment accurately and logically.

Teaching and learning materials: Rulers, compasses, protractors for accurate graphing, Bank rate brochures for interest calculations, Graph papers, Scientific calculators and software: (GeoGebra, Desmos, Excel)or AI.

Topic Area: Geometry			Sub-topic Area: Space geometry	
S5 Mathematics		UNIT 4: Points, straight lines and shapes in 2D		Number of periods: 14
Key Unit Competence: Apply the meaning of coordinates in solving problems involving points, lines and shapes in 2D space.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain key concepts: Cartesian coordinates, vector/parametric equations, and geometric shapes. – Apply the formulas for distance, midpoint, angles, and equations of lines and circles.	– Represent points, lines and circles, using Cartesian, vector, and parametric forms. – Convert between different forms of linear equations (Cartesian, vector, parametric). – Calculate distances, midpoints, angles in 2D.	– Appreciate the power of coordinate systems in algebra and geometry. – Value precision and accuracy in formulating equations and performing calculations.	– Points in 2D: Cartesian coordinates of a point, Distance between two points, Mid-points of a line segment – Lines in 2D: Cartesian equation of a line, Vector equation of a line, Parametric equations of a line	– 2D graphing lab: Use graphing software (e.g., GeoGebra, Desmos) to visually explore how changes in equations affect graphs of lines, shapes and circles. Students match given geometric conditions to their correct equations.

– Identify the different forms of linear equations and their interrelationships. – Describe the positional relationships between points, lines, and circles.	– Determine the position of points relative to lines and circles, (e.g., inside/ outside). – Analyze the relationship between lines (parallel, intersecting). – Solve complex geometric problems involving collinearity, and shortest distances. – Model real-world situations (e.g., navigation, physics) using coordinate geometry.	– Develop spatial reasoning and visualization skills for 2D problems. – Collaborate effectively to tackle multi-step geometric challenges.	– Problems on points and straight lines in 2D: Position of a point to a line – Problems on points and straight lines in 2D: Angle between two lines – Problems on points and straight lines in 2D: Distance from a point to a line. – Shapes in 2D and coordinates. – Definition and Cartesian equation of a circle – Problems involving position of a circle and a point in 2D.	– In teams, solve a series of interconnected 2D coordinate problems (e.g., “Find the distance from point A to the line, then use that as a radius to define a circle”). – Given points in 2D, students use multiple methods, such as the dot product or systems of equations, to determine if the points are collinear or coplanar.
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				– Circle scenarios: Students analyze different positional cases between a circle and a plane/line (intersecting, tangent, no intersection) and present the algebraic conditions that lead to each case.
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Link to other subjects: Physics *(Trajectory of projectiles (parabolic motion defined in a plane, electric fields around a point charge (spheres), and force components.*

Assessment criteria: *Students apply algebraic and geometric concepts to solve problems involving distances, angles, and positional relationships in 2D contexts accurately.*

Teaching and learning materials: *Graphing calculators, dynamic geometry software (GeoGebra) or AI, physical modeling kits (straws, clay, string), graph paper and poster paper, real-world problem sets.*

Topic area: Analysis			Sub-topic area: Limits, differentiation and integration	
S5 Mathematics		UNIT 5: Trigonometric and inverse trigonometric functions.		Number of periods: 35
Key Unit Competence: Use trigonometric and inverse trigonometric functions in solving mathematical and real-life problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain trigonometric functions and their inverses. – State and describe the domain and range of six trigonometric functions and their inverses.	– Compute and sketch graphs showing the domain, range, and period of trigonometric and inverse trigonometric functions. – Determine parity and periodicity from trigonometric equations and graphs.	– Develop precision, patience, and logical reasoning when solving	– Introduction on trigonometric functions, their inverses and related periods. – Domain and range of six trigonometric functions and inverses for trigonometric functions. – Range of inverses of trigonometric functions	– Presents the six trigonometric functions and their inverses using real-life examples (angles, rotations, waves). – -Work in pairs to identify and discuss trigonometric functions and their inverses, and practice converting between them using guided exercises.

– Explain the concepts of parity and periodicity in trigonometric functions. – Identify and determine limits of trigonometric and inverse trigonometric functions. – Recall and apply rules for derivatives of sine, cosine, tangent, cotangent, secant, and cosecant functions. – State and derive formulas for the derivatives of inverse trigonometric functions.	– Evaluate limits involving trigonometric and inverse trigonometric functions. – Differentiate trigonometric and inverse trigonometric functions accurately. – Find successive derivatives of sine, cosine, and other trigonometric functions. – Apply derivatives of trigonometric functions to solve practical problems in refraction, simple harmonic motion, and optimization.	– trigonometric problems. – Show curiosity and critical thinking in exploring patterns of trigonometric derivatives. – Appreciate the importance of trigonometric models in understanding physical phenomena and real-world applications.	– Parity of trigonometric functions – Limits of trigonometric and inverse of trigonometric functions. – Derivative of sine and cosine – Successive derivative for sine and cosine. – Derivative of tangent and cotangent. – Derivative of secant and cosecant. – Derivative of inverse sine, inverse cosine, inverse tangent and inverse cotangent, inverse secant and inverse cosecant.	– Demonstrate how to determine domain and range of sine, cosine, tangent, cotangent, secant, and cosecant and their inverses and students sketch graphs of these functions and identify domain and range from the graphs the complete exercises on finding domain and range. – Demonstrate the evaluation of limits using identities and standard formulas, after which solve guided exercises individually and in pairs, explaining their reasoning and results.
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– Explain the meaning of successive derivatives and their patterns. – Recognize real-life phenomena modeled by trigonometric functions.	– Interpret and communicate results of trigonometric models in real contexts.	– Demonstrate collaboration, consistency, and a positive attitude during group and individual tasks.	– Successive derivatives. – Guided practice on Successive derivatives. – Applications of trigonometric functions: refraction of light in a prism, simple harmonic motion problems and optimization.	– Present differentiation rules with step-by-step derivation for sine, cosine, tangent, cotangent, secant, cosecant, and their inverses. – Practice differentiation exercises, verify results with calculators, software or AI and discuss patterns. – Compute successive derivatives of all trigonometric functions, identify patterns, and present results in groups.
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Link to other subjects: Physics (Modeling simple harmonic motion, Refraction of light and optics problems, Wave motion and oscillations).

Assessment criteria: Students identify trigonometric functions and their inverses, find their domains, ranges, and derivatives, and use these ideas to solve real-world problems such as wave motion, light refraction, and optimization.

Teaching and learning materials: Rulers, compasses, and protractors for accurate graphing, Computers, tablets, or smartboards with graphing software (e.g., GeoGebra, Desmos, Excel) or AI.

Topic area: Linear Algebra			Sub-topic area: Linear transformation in 3D	
S5 Mathematics		UNIT 6: Matrices and determinants of order 3		Number of periods: 21
Key Unit Competence: Apply properties of matrices, determinant of order 3 and operations on linear transformations in solving problems in R^3				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain square matrices of order 3 and their types (diagonal, scalar, identity, zero, symmetric, etc.). – Recall the rules and properties of matrix addition, subtraction, multiplication, and transpose.	– Perform addition, subtraction, and multiplication of 3×3 matrices correctly. – Compute the transpose of a matrix and verify its properties. – Calculate the determinant of a 3×3 matrix.	– Demonstrate accuracy, precision, and logical reasoning in computations. – Appreciate the usefulness of matrices in modeling and solving real-life problems.	– Introduction on square matrices of order 3 – Types of matrices – Addition on matrices and properties – Subtraction on matrices and properties – Transpose of matrices and properties	– Presents the meaning and examples of 3×3 matrices and their types (diagonal, scalar, identity, zero, symmetric) and identify types of matrices from given examples and discuss their characteristics in pairs, then complete guided exercises classifying matrices by type.

– Explain the concept of matrix inverse and its properties. – Demonstrate understanding of determinants of order 3 and their properties. – Recognize the application of matrices in solving systems of linear equations from real-life problems.	– Determine the inverse of a matrix and apply its properties. – Solve systems of three linear equations using matrices and interpret solutions in real-life contexts. – Apply matrices to model and solve practical problems.	– Develop persistence and analytical thinking in solving matrix-related problems.	– Multiplication of matrices and Properties – Matrix of linear transformation in 3 dimensions – Determinants of order 3 and properties – Matrix inverse and Properties – Application of matrices in everyday life.	– Practice addition, subtraction, and multiplication through individual exercises and verify the properties using calculations or tools such as GeoGebra, or MATLAB. – Practice evaluating determinants of 3×3 matrices and explore how properties simplify calculations. – Compute inverses of 3×3 matrices, verify properties, and discuss implications. – Group discussions to reflect on the usefulness of matrices in solving real-world problems
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Link to other subjects: *Physics*(forces, equilibrium, solving electrical circuit problems and network flows), *Computer Science*(applying matrices in 3D graphics, animation, and image processing), *Economics / Business*(modeling input-output systems, budgeting, and resource allocation), *Biology* (modeling population interactions or gene distribution), *Chemistry* (balancing chemical equations, representing stoichiometric relationships and reaction networks).

Assessment criteria: Students apply matrices and determinants of order 3 in modeling, solving, and interpreting real-life problems accurately.

Teaching and learning materials: Flashcards with different types of matrices and their properties, Charts showing rules of matrix operations and determinants, graphing software or dynamic math tools (e.g., GeoGebra) or AI.

Topic area: Linear Algebra			Sub-topic area: Linear transformation in 3D	
S5 Mathematics		Unit 7: Vector space of real numbers		Number of periods: 28
Key Unit Competence: Apply operations of vectors in three dimensions in solving mathematical and real-life problems involving magnitude, direction, work, area, and volume				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain vectors, scalar (dot) product,vector (cross) product, and mixed product. – State the properties of vector operations in 3D. – Identify sub-vector spaces, linear combinations, and bases.	– Represent vectors geometrically and algebraically in 3D. – Perform operations on vectors (addition, scalar multiplication, dot and cross products).	– Value the power of vector notation in simplifying complex spatial problems. – Appreciate the interplay between geometric intuition and algebraic computation. – Develop perseverance and accuracy in vector calculations.	– Introduction to the Vector space R^3 : vector position in R^3 – Operation of vectors in 3 dimensions and their properties – Properties of operation of vectors in 3D – Sub-vector spaces	– In groups, students use physical models (sticks, strings, cardboard) or 3D software (like GeoGebra) to visualize vector addition, subtraction, and the geometry of dot and cross products.

– Recall formulas for magnitude, the angle between vectors, work, area, and volume.	– Calculate the magnitude of a vector, the angle between two vectors, and direction cosines. – Determine if a set of vectors is linearly independent, spans a space, or forms a basis. – Compute the coordinate vector relative to a given basis. – Apply dot and cross products to calculate work done, areas of parallelograms/ triangles, and volumes of parallelepipeds/ tetrahedrons.	– Collaborate effectively in hands-on and problem-solving activities.	– Linear combination of vectors – Basis and dimension – Coordinate vector relative to the given base – Scalar or Dot product of two vectors and properties – Magnitude (or norm or length) of a vector – Angle between two vectors and the direction cosines – Vector product and its properties	– Basis Hunt": Given different sets of vectors in 3D, students determine which sets form a basis for R^3. They then find the coordinates of a known vector relative to a new, non-standard basis. – Work and Energy lab: Students calculate the work done by a force (e.g., pulling a object at an angle) using $W = \vec{F} \cdot \vec{dW} = F \cdot d$ – They compare scenarios where the force is parallel, perpendicular, or at an angle to the displacement.
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	– Solve real-world problems involving force, displacement, and spatial geometry.		– Mixed product – Applications of scalar and vector products: Work done by the force – Area of a parallelogram – Volume of a parallelepiped and tetrahedron.	– Groups are tasked with calculating the area of an irregular plot (as a parallelogram) or the volume of a slanted building (as a parallelepiped) using vector products. They present their solutions. – Vector Product Properties Investigation: Students experimentally verify the properties of the cross product (e.g., anti-commutativity, distributivity) using specific vector examples and then justify them algebraically.
Link to other subjects: Physics: Physics(Force, displacement, work, torque , electric and magnetic fields);Geography(Navigation using direction cosines and displacement vectors).				

Assessment criteria: *The students apply vector operations and related concepts in solving problems involving work, area, volume, and direction in both abstract and real-life contexts logically.*

Teaching and learning materials: *Markers, GeoGebra (or similar 3D graphing software), AI, physical models for vectors (arrows, protractors), force sensors and carts for lab activity, worksheets, textbooks, poster paper for presentations.*

Topic area: Geometry			Sub-topic Area: Space geometry	
S5 Mathematics		UNIT 8: Points, straight lines and shapes in 3D.		Number of periods: 28
Key Unit Competence: Apply the meaning of coordinates in solving problems involving points, lines, planes and shapes in 2D and 3D space.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain key concepts: Cartesian coordinates, vector/parametric equations, and geometric shapes (sphere, plane). – Apply the formulas for distance, midpoint, angles, and equations of lines, planes, and spheres.	– Represent points, lines, shapes, and spheres using Cartesian, vector, and parametric forms. – Convert between different forms of linear equations (Cartesian, vector, parametric).	– Appreciate the power of coordinate systems in algebra and geometry. – Value precision and accuracy in formulating equations and performing calculations.	– Cartesian coordinates of a point in 3D. – Distance between two points in 3D. – Coordinates of a midpoint of a segment and centroid of a geometric figure – The ratio formula	– 3D graphing lab: Use graphing software (e.g., GeoGebra, Desmos) to visually explore how changes in equations affect graphs of lines, circles, and planes. Students match geometric conditions to their correct equations. – In teams, solve a series of interconnected and 3D coordinate problems (e.g., “Find the distance from point A to the line, then use that as a radius to define a sphere...”).

– Identify the different forms of linear equations and their interrelationships. – Describe the positional relationships between points, lines, planes, and spheres.	– Calculate distances, midpoints, angles, and centroids in 3D. – Determine the position of points relative to lines, planes, and spheres (e.g., inside/outside). – Analyze the relationship between lines (parallel, intersecting, skew) and planes (parallel, intersecting)	– Develop spatial reasoning and visualization skills for 3D problems. – Collaborate effectively to tackle multi-step geometric challenges.	– Equations of lines in 3 dimensions – Condition of co-linearity of 3 points – Relationships between lines – Angle between two lines – Distance from a point to a line – Shortest distance between two skew lines – Equations of planes in 3 dimensions – Condition of coplanarity of four points – Position of a line and a plane – Angles between lines and planes	– Architectural Blueprint Project: Students act as architects/engineers. Given specific points and constraints, they must derive equations for roof lines (3D lines), determine if structural beams (lines) will intersect, and calculate the shortest supporting wire (distance from point to line). – 3D model construction: Using straws, string, and modeling clay, groups build physical models to represent skew lines, intersecting planes, and spheres. They then use coordinate geometry to verify their models' properties (angles, distances).
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	– Solve complex geometric problems involving collinearity, coplanarity, and shortest distances. – Model real-world situations (e.g., navigation, architecture, physics) using coordinate geometry.		– Shortest distance from a point to a plane – Position of planes – Equation of a sphere in 3 dimensions – Position of point and sphere – Position of a sphere and a line – Position of a sphere and a plane. – Position of two spheres.	– Given four points in 3D, students use multiple methods, such as the vector cross product or systems of equations, to determine if the points are coplanar. If not, they calculate the volume of the resulting tetrahedron, then, use AI tools or GeoGebra for self-correction. – Sphere scenarios: Students analyze different positional cases between a sphere and a plane/line (intersecting, tangent, no intersection) and present the algebraic conditions that lead to each case.
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Link to other subjects: **Physics** (Trajectory of projectiles (parabolic motion defined in a plane, electric fields around a point charge (spheres), and force components); **Geography and GIS** (Global Positioning Systems (GPS) use 3D coordinates on a spherical model of the Earth); **Robotics** (Path planning for robot arms involves calculating angles and distances between points in 3D space).

Assessment criteria: Students apply algebraic and geometric concepts to solve problems involving distances, angles, and positional relationships in 3D contexts accurately.

Teaching and learning materials: Graphing calculators, dynamic geometry software (GeoGebra, Desmos 3D) or AI, physical modeling kits (straws, clay, string), graph paper and poster paper, real-world problem sets.

Topic area: Statistics and probability			Sub-topic area: Descriptive statistics	
S5 Mathematics		UNIT 9: Bivariate statistics.		Number of periods: 14
Key Unit Competence: Apply correlation and regression analysis in interpreting bivariate data and making predictions.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain bivariate data and distinguish it from univariate data. – Identify and describe different types of correlation from a scatter diagram (positive, negative, no correlation, strong, weak).	– Collect and organize bivariate data into a frequency distribution table. – Construct and interpret scatter diagrams to visually assess the relationship between two variables.	– Appreciate the power of statistical analysis in revealing hidden relationships in real-world data. – Develop a critical attitude towards data and statistical claims, especially regarding causation.	– Introduction to bivariate data – Definition and examples of bivariate data. – Scatter diagram – Types of correlation – Frequency distribution table of bivariate data	– Data collection and visualization: – In small groups, students collect their own bivariate data (e.g., weight vs. height, study time vs. distance). – Use graph paper or software such as excel to plot scatter diagrams of collected and provided datasets. – Classify the type and strength of correlation observed. – Calculation and analysis: – Guided practice in creating bivariate frequency tables.

– State the formulas for covariance, Pearson's correlation coefficient, and the equations of regression lines. – Explain the meaning and properties of the correlation coefficient (r) and the coefficient of determination (r^2). – Understand the concept of a line of best fit.	– Calculate the covariance and Pearson's product-moment correlation coefficient. – Determine the equation of the least-squares regression line (y on x and x on y). – Use the regression line to make predictions (interpolation and extrapolation) and understand their limitations. – Critically interpret statistical results in context, avoiding confusion between correlation and causation.	– Foster accuracy and precision in calculations and graphical representations. – Value the collaborative nature of data collection and analysis.	– Covariance – Line of the best fit – Regression lines – Coefficient of correlation. – Coefficient of determination (r^2). – Applications: Data analysis, interpretation and prediction problems in various areas.	– Step-by-step calculation of covariance, correlation coefficient (r), and the regression line equation using a worked example. – “Prediction Game”: Apply regression equations to predict values and verify their accuracy using scientific calculator or excel sheets for immediate feedback. – Real-world application: – Case Study Analysis: Examine real or simulated data from business (advertising cost vs. sales), geography (altitude vs. temperature), or biology (Weight vs height). – Group presentations interpreting the statistical output (r , r^2 , regression equation) and its meaning in the context of the case study.
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Link to other subjects: **Economics** (Demand/supply curves, analyzing relationships like price and quantity demanded); **Geography** (Climate data analysis (e.g., rainfall and crop yield)); **Biology** (Relationship between variables like plant growth and sunlight/fertilizer); **Physics** (Relationships in experimental data (e.g., force and acceleration in Hooke's law)).

Psychology (Analyzing survey data between two variables).

Assessment criteria: The student accurately constructs a scatter diagram from given bivariate data, identifies and describes the type and strength of correlation displayed on a scatter diagram, calculates the correlation coefficient and the equation of a regression line, interprets the value of the correlation coefficient and the slope of the regression line in the context of a given problem, uses the regression line equation to make sensible predictions and interprets the results.

Teaching and learning materials: Graph paper, rulers, scientific calculators. Spreadsheet software (e.g., Excel, Google Sheets, GeoGebra) or AI for demonstration and student practice. Prepared datasets on worksheets or digital files. Chart posters or digital slides showing examples of different scatter diagrams. Real-world case study handouts from various subjects.

Topic area: Statistics and probability			Sub-topic area: Probability	
S5 Mathematics		UNIT 10: Elementary and Conditional Probability.		Number of periods: 28
Key Unit Competence: Apply counting techniques and probability rules in determining and interpreting the likelihood of events under equally likely assumptions.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain key terms: experiment, outcome, sample space, event (simple and compound). – Understand and state the three basic axioms of probability. – Explain mutually exclusive (disjoint) and independent events.	– Calculate the probability of an event using the classical (theoretical) approach. – Apply the addition rule for mutually exclusive and non-mutually exclusive (inclusive) events. – Apply the multiplication rule for independent and dependent events.	– Appreciate the role of probability in quantifying uncertainty and making informed decisions in daily life.	– Introduction to probability in real life. – Concept of probability (the probability, experiment, sample space and event) and examples. – Probability of an event.	– Foundational activities: – “Probability in a Bag”: Hands-on activity with colored counters/ marbles to introduce basic concepts. – Class discussion: Explore real-life examples of probability (weather forecasts, game chances, insurance) and discuss how probability informs decision-making in each case.

– Explain the difference between independent and dependent events. – Understand the concept of conditional probability as a change in the sample space. – Apply the formulas for the addition rule, multiplication rule, and conditional probability. – Understand the purpose and application of Bayes' Theorem.	– Construct and use tree diagrams to model multi-stage experiments and calculate probabilities. – Calculate conditional probabilities using the formula $P(B A) = P(A \text{ and } B) / P(A).$ – Use two-way tables and Venn diagrams to organize information and solve probability problems. – Apply Bayes' Theorem to solve practical problems involving conditional probabilities: $P(A B) = [P(B A) \times P(A)] / P(B).$	– Cultivate a critical attitude towards probabilistic claims in media, gambling, and risk assessment. – Foster collaborative skills through group problem-solving activities. – Value accuracy in calculation and clarity in the interpretation of results.	– Basic probability rules. – Probability of mutually exclusive (incompatible) events. – Probability of inclusive events and additive rule. – Concept of independent events and examples. – Probability of independent events. – Multiplication rule. – Multiplication rule using tree diagrams. – Tree diagrams and probability problems.	– “Sample Space Race”: Groups compete to list all possible outcomes for given experiments (e.g., rolling two dice). – Rule application and visualization: – Use Venn diagrams and two-way tables to visualize and solve problems on mutually exclusive and inclusive events. – “Tree Diagram Challenge”: Model complex scenarios (e.g., a best-of-three sports series, manufacturing defects) using tree diagrams. – “Are they Independent?": Activities to test for independence using the rule $P(A \text{ and } B) = P(A) \times P(B)$.
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	– Interpret probability values in context, assessing likelihood and risk.		– Introduction to dependent events. – Conditional probability: Probability of event B occurring when event A has already taken place. – Basic formulae and properties of conditional probability. – Bayes theorem: $P(A B) = [P(B A) \times P(A)] / P(B).$ – Application of Bayes' theorem.	– Case Studies: Analyze simple legal, business, or diagnostic scenarios where Bayes' Theorem helps revise beliefs with new evidence. – Explore videos with guided derivations and worked examples for conditional probability and Bayes' formula.
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Link to other subjects: **Biology:** Genetics (inheritance patterns), epidemiology (disease spread and testing). **Business:** Risk analysis, actuarial science, and quality control. **Economics:** Forecasting and market analysis. **Geography:** Risk assessment for natural disasters. **Sports Science:** Game strategies and performance predictions.

Assessment criteria: A student accurately define core probability concepts; correctly calculate probabilities for simple, mutually exclusive, and inclusive events; successfully apply multiplication rules for independent and dependent events, utilizing tools like tree diagrams; compute and interpret conditional probabilities to solve problems; apply Bayes' Theorem in practical contexts; and, crucially, interpret and communicate the meaning of their calculated probabilities within the context of a real-world scenario.

Teaching and learning materials: Bags, colored counters, dice, coins, and playing cards for hands-on experiments. Scientific calculators. Graph paper for drawing tree diagrams and Venn diagrams. Prepared worksheets with problem sets and case studies. Digital display for demonstrating solutions and simulations, AI.

SENIOR SIX

2.4 Mathematics program for secondary six

2.4.1 Key competences at the end of secondary six

After completion of secondary 6, the mathematics syllabus will help the learner to:

- Apply properties of operations of complex numbers in solving mathematical problems.
- Solve polynomial equations in the set of complex numbers and solve related problems in physics.
- Extend the use of concepts and definitions of functions to determine the domain of logarithmic and exponential functions.
- Use integration as the inverse of differentiation and as the limit of a sum and apply them to finding area and volumes to solve various practical problems.
- Use differential equations to solve related problems that arise in a variety of practical contexts.
- Relate the sum and the intersection of subspaces of a vector space by the dimension formula.
- Determine the kernel and the image of a linear transformation and use the results to infer the properties of a linear transformation.
- Transform a matrix to its equivalent form using elementary row operations.
- Determine algebraic representations of conics in the plane and use them to represent and interpret physical phenomena.
- Explore the parameters of a random variable (discrete or continuous) for a given context.

2.4.2 Mathematics units for secondary six

Topic area: Algebra			Sub-topic area: Numbers And Operations	
S6 Mathematics		Unit 1:Complex numbers		Number of periods: 35
Key Unit Competence: Apply properties for operations of complex numbers in solving mathematical problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe a complex number and the imaginary unit i. – Identify the real and imaginary parts of a complex number. – State the properties of complex number operations (commutative, associative, distributive).	– Represent complex numbers on the Argand plane. – Perform arithmetic operations (addition, subtraction, multiplication, division) with complex numbers.	– Appreciate the usefulness of complex numbers by recognizing how algebraic, polar, and exponential forms simplify problem-solving and reveal the connection between algebra and trigonometry.	– Basic concepts of complex numbers: • Introduction to the concept of complex numbers • Definition and properties of the imaginary unit (i). – Representation of complex numbers and their properties • Geometric representation of complex numbers on the Argand plane. • Modulus and argument • Equality of complex numbers	– Using graph paper or GeoGebra, students plot complex numbers like $2 + 3i$ on the Argand plane and identify their modulus and argument. – In pair, students perform addition, subtraction, multiplication, and division of complex numbers, such as $(2 + 3i)(1 - i)$.

– Recall the definitions of modulus, argument, and conjugate. – State De Moivre's Theorem and Euler's Formula. – Understand the geometric interpretation of complex number operations.	– Calculate the modulus and argument of a complex number. – Convert between rectangular, polar, and exponential forms. – Solve linear and quadratic equations in the complex plane.	– Demonstrate confidence in applying the properties and formulas of complex numbers such as De Moivre's and Euler's formulas when transforming and solving mathematical expressions and equations.	– Operations with Complex Numbers • Addition and subtraction of complex numbers. • Conjugate and opposite of a complex number. • Multiplication of complex numbers • Inverse and division of complex numbers • Square root of a complex number – Equations and polynomials in the set of complex numbers: • Linear equations in the set of complex numbers. • Quadratic equations in the set of complex numbers. • Polynomials in the set of complex numbers.	– Find the conjugate, opposite, and inverse of a complex number, and explore geometrically how conjugation reflects a point across the real axis. – Solve linear and quadratic equations with complex roots, for example solving $x^2 + 4 = 0$. – Use calculators or software to verify solutions and plot them on the Argand plane, observing symmetry in conjugate pairs.
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	– Find the n-th roots of a complex number. – Apply De Moivre's Theorem to derive trigonometric identities. – Model simple real-world problems (e.g., AC circuits) using complex numbers.		– Geometric representation and loci: • Argument of a complex number. • Loci related to angles. • nth roots of a complex number and its graphical representation. • Construction of regular polygons. – Complex numbers in polar and exponential form: • Polar form of a complex number • Multiplication and division in polar form • Powers of complex numbers in polar form • De Moivre's formula. • Exponential form of a complex number • Euler's formula of complex numbers.	– Think and pair: Students explore loci of complex numbers using rulers and compasses, such as drawing $z - 2 = 3$ to form a circle. – Students are guided to use any a video or GeoGebra in visualizing the argument and nth roots of unity and construct regular polygons representing these roots. – Convert between algebraic and polar forms, for example: $+i = \sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$
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			– Applications: • Product-to-sum formulas in trigonometry using complex numbers. • Trigonometric form of a multiple of an angle. • Linearization of trigonometric expressions. • Solving equations of the form $a \cos \theta = b$. • Simple alternating current problems.	– Apply De Moivre's and Euler's formulas to find powers and products, such as $e^{in\theta}$, and visualize the effect as rotation and scaling on the Argand plane. – In group, use complex numbers to derive and apply trigonometric formulas, such as expressing $\cos A \cos B$ using exponential forms. – Simplify trigonometric expressions and solve related equations involving complex representations.
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				– Students apply complex numbers to model alternating current, for example representing $I = I_0 e^{i\omega t}$ in AC circuits.
Link to other subjects: Physics (Alternating current and mechanical oscillations)				
Assessment criteria: Students perform operations on complex numbers in different forms and use complex numbers to solve related real life problems involving geometry, trigonometry, and alternating currents.				
Teaching and learning materials: Scientific calculators, GeoGebra or Desmos, PhET simulations, charts and posters and any AI.				

Topic area: Analysis			Sub-topic area: Limits, Differentiation And Integration	
S6 Mathematics		Unit 2: Logarithmic And Exponential Functions.		Number of periods: 28
Key Unit Competence: Extend the concepts of functions to investigate logarithmic and exponential functions and their applications in real life.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe the properties, domain, range, limits, and continuity of logarithmic and exponential functions. – Explain derivatives, variations, and curves of logarithmic and exponential functions.	– Illustrate logarithmic and exponential functions. – Use differentiation techniques to find the derivatives of logarithmic and exponential functions.	– Appreciate the usefulness of exponential and logarithmic models in solving real-life problems. – Develop accuracy and logical reasoning in interpreting logarithmic and exponential curve.	– Natural logarithmic functions: • Domain, range, Limit, continuity and asymptotes of natural logarithmic function • Derivative, variation and curve of natural logarithmic function – Logarithmic functions with any base: • Domain, range, limit, continuity and asymptotes logarithmic function with any base	– Interactive discovery: Use calculators and plotting software (GeoGebra) to explore the shape of $y = \ln x$ and $y = \log_a x$ for different bases. – Pair work: Students determine domain and range experimentally by entering negative and zero values. –

– Recognize and interpret asymptotes of these functions. – Recognize real-life situations modelled by logarithmic and exponential functions.	– Sketch and analyse curves showing growth and decay behaviour. – Apply logarithmic and exponential functions to solve real-world problems involving change over time.	– Show curiosity and confidence in applying logarithmic and exponential function to everyday contexts.	• Derivative, Variation and curve of logarithmic function with any base – Exponential function with base e . • Domain, range, limit, continuity and asymptotes of exponential function with base e • Derivative, variation and curve of exponential function with base e – Exponential function with any base • Domain, range, limit, continuity and asymptotes of exponential function with any base. • Derivative, variation and curve of exponential function with any base.	– Class discussion: Students identify asymptotes and explain why logarithmic functions are undefined for $x \leq 0$. – Hands-on plotting: Learner's plot $y = e^x$, $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ to compare growth and decay. – In groups, students explore tables of values to determine limits and asymptotes. – Real-life projects: Students conduct research on problems on compound interest, population growth, and depreciation using exponential formulas, solve them and present findings.
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			– Applications of logarithmic and exponential functions: • Compound interest, Mortgage amount and depreciation value problems. • Population growth problems. • Earthquake problems. • Radioactive decay problems. • Carbon-14 dating. • Problems about alcohol and risk of car accidents.	– Experiment or simulation: Model alcohol elimination using $B(t) = B_0 e^{-kt}$ and use logarithms to find safe driving time. – Group task: Compare exponential growth in population with exponential decay in radioactivity. – Case study: Interpret logarithmic scales in earthquake magnitude and carbon dating. – Class discussion: Reflect on how logarithmic and exponential functions are used to model and inform responsible behaviour about alcohol and driving.
Link to other subjects: Physics (Newton's law of cooling), Geography (magnitude of an Earthquake, population growth), Economics (compounded interest, mortgage depreciating value), Biology (population growth, alcohol elimination), Chemistry (carbon dating).				

Assessment criteria: Students investigate logarithmic and exponential functions, find the domain of definition, the limits, asymptotes, variations, graphs and model related real life problems.

Teaching and learning materials: Cutouts with different logarithmic or exponential functions, Graph paper, GeoGebra, and any AI.

Topic area: Analysis		Sub-topic area: Limits, Differentiation And Integration		
S6 Mathematics		Unit 3: SERIES		Number of periods: 14
Key Unit Competence: Apply expansion on series in solving problems about approximations and limits of some mathematics expressions.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe a power series and the concepts of convergence and divergence. – State the general form of a Taylor series about a point $x=a$ and a Maclaurin series (as a special case where $a=0$).	– Differentiate finites from infinite series. – Determine the radius and interval of convergence for a given power series. – Calculate the derivatives of a function to a specified order. – Construct Taylor and Maclaurin series for a given function.	– Appreciate the role of series in simplifying complex calculations. – Develop accuracy and logical reasoning in mathematical derivations.	– Generalities on series: Finite and infinite series, power series, convergence of series – Expansion on series: • Taylor series • Maclaurin series	– Students brainstorm examples of sequences and extend them to series. Explore their convergence and divergence through graphical and numerical examples. – In pairs, find Taylor and Maclaurin series of common functions (e.g., $\sin x$, $\cos x$, e^x, $\ln(1+x)$). – Students use GeoGebra or other graphing tools to visualize the accuracy of series approximations. – Students are guided to solve exercises on approximations and limits using Taylor series. e.g: $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

– Recall the standard Maclaurin series. – Understand the concept of using a polynomial to approximate a function and the role of the remainder term. – Explain how series can be used to represent irrational numbers and estimate a natural logarithm number and roots of equations.	– Apply properties of series to solve other mathematical problems: the number e, number π, estimation of trigonometric number of an angle, estimation of an irrational number, Calculation of limits, estimation of a natural logarithm number and estimation of roots of equations.	– Show curiosity and confidence in applying Taylor's series to solve other problems.	– Applications of series: • Estimation of the number e • Estimation of the number π • Estimation of trigonometric number of an angle. • Estimation of an irrational number. • Calculation of limits • Estimation of a natural logarithm number. • Estimation of roots of equations.	– In small groups, students discuss applications of series (for example, a position $x(t)$ under constant acceleration can be expressed as a Taylor series around time $t = 0$).
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Link to other subjects: Physics (Approximations in Classical Mechanics).

Assessment criteria: Students correctly derive and explain Taylor and Maclaurin series for standard functions and apply them accurately to approximate values, solve limits, and integration problems

Teaching and learning materials: Graphing calculators, GeoGebra software, charts showing function expansions and any AI.

Topic area: Analysis		Sub-topic area: Limits, Differentiation And Integration		
S6 Mathematics		Unit 4: Integration		Number of periods:49
Key Unit Competence: Apply integration as the inverse of differentiation and as the limit of a sum in determining areas of plane surfaces, volumes of solids of revolution, and lengths of curved lines.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the concept of integration as the inverse of differentiation and as the limit of a sum – Describe the properties and rules of indefinite and definite integrals.	– Compute differentials, indefinite and definite integrals using various techniques. – Apply integrals to determine areas, volumes, and physical quantities.	– Appreciate integration as a powerful tool in mathematical modelling and real-world problem-solving. – Develop persistence, accuracy, and logical reasoning when solving integration problems.	– Differentials. – Indefinite integrals: Definition and properties – Methods of Integration: By substitutions, Integration by parts	– Concept building: Through guided discussion, students identify integration as the inverse of differentiation by performing both processes on simple polynomial functions to observe the relationship between them. – Graphical approach: Students use GeoGebra, graphing calculators to divide a curve into consecutive subintervals and calculate the areas of corresponding rectangles. By observing how the sum of these areas changes as the width of each subinterval tends to zero.

– Identify appropriate methods of integration for various functions.	– Analyse and solve problems involving integrals with accuracy and logical reasoning.	– Show curiosity, confidence, and teamwork in applying integration techniques.	– Integration of rational function: Numerator is expressed in terms of derivative of denominator, Numerator is greater or equal to the degree of denominator, Denominators are factorized into linear factors, denominator containing linear and quadratic. – Integration of trigonometric functions: $\int \sin mx \cos nx \, dx,$ $\int \cos mx \cos nx \, dx,$ $\int \sin mx \sin nx \, dx$	– Students are guided by teacher to develop the concept of integration as the limit of a sum. They visualize the area under a curve, verify analytical results, and interpret integration as the inverse of differentiation – Practical geometry connection: Students recall and calculate the areas of simple plane figures such as the square, rectangle, triangle, trapezium, and circle, then extend this to determine the volumes of solids such as a sphere, cylinder, and cone, and the circumference of a circle. – Working in small groups, students derive and apply formulas for the main integration methods: substitution and integration by parts through analysis of worked examples and completion of missing steps.
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			• form $\int \cos^m x \sin^n x dx$ • form $\int \tan^m x \sec^n x dx$ • Involving $\sin x$, $\cos x$, $\tan x$ on denominator. • Involving $\sin^2 x$, $\cos^2 x$ on denominator. – Integral of irrational function: • Involving $\sqrt[n]{ax+b}$ • Involving $\sqrt{ax^2+bx+c}$ – Integration by reduction formulae.	– Students then extend this understanding to integrate rational functions, especially cases where the numerator is the derivative of the denominator or when its degree is greater or equal to that of the denominator, after factorizing denominators into linear or quadratic forms. – Students apply integration to determine areas of plane surfaces and the volumes of solids of revolution using disk, washer, and shell methods, and to solve physics-related problems such as work done, displacement, and fluid flow.
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			– Integration by Maclaurin's series. – Definite integrals: Definition, properties and techniques for definite integrals. Application of integrals: – Calculation of area of plane surfaces – Calculation of volume of a solid of revolution by disc method. – Calculation of volume of a solid of revolution for two defined functions (Washer method).	
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			– Calculation of volume of a solid of revolution by using "The Shell method" – Calculation of the arc length of a curved line After – Application of definite integrals in physics and finance.	
Link to other subjects: Physics (Applications of integration in motion, work, energy, and fluid mechanics), Engineering (Calculation of surface areas and volumes in topographic analysis).				
Assessment criteria: Students accurately compute indefinite and definite integrals using appropriate techniques and apply them in determining areas, volumes, and related real-life situations.				
Teaching and learning materials: Mathematical tables and formula sheets; Graphing calculators, GeoGebra software, and any AI.				

Topic area: Analysis			Sub-topic area: Limits, Differentiation And Integration	
S6 Mathematics		Unit 5: Differential Equations		Number of periods:28
Key Unit Competence: Apply ordinary differential equations of first and second order in solving real life problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe and classify ordinary differential equations.	– Solve first and second order differential equations using appropriate methods. – Apply differential equations to model physical, biological, and economic problems.	– Appreciate the role of differential equations in modelling real-world phenomena. – Show curiosity and confidence when applying differential equations to practical situations.	– Introduction to differential equations: Definition and classification. – First order differential equations: • Differential equations with separable variables • Simple homogeneous equations	– In groups or individually, solve differential equations, experiment with varying initial conditions and parameters to see how solutions change over time. Problem-solving in physics: Students formulate differential equations for Newton's law of cooling and simple electrical circuits (RC and RL circuits).

– Identify first and second order differential equations and their types.	– Interpret solutions of a differential equation in real life context.		• Linear differential equations – Particular solution of a differential equation – Second order differential equations: Meaning, constant coefficients: (two distinct real roots, double root, complex roots). – Non-homogeneous equations with right-hand sides: • $r(x) = pe^{ax}$ • $r(x) = pe^{ax} \cos \beta x + Qe^{ax} \sin \beta x$ • Sum of two or more functions	– Students solve these equations analytically and numerically, compare solutions with graphs, and discuss physical interpretations, such as how resistance, capacitance, or cooling rate affects outcomes. – Students model population growth and decay using first-order differential equations and extend to logistic growth models to study carrying capacity. – Students compare methods such as separation of variables, integrating factor method, and characteristic equation approach. -Students solve similar problems using multiple methods to understand the advantages and limitations of each method.
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			– Application of differential equations: • Newton's law of cooling, electrical circuits (RC, RL circuits). • Population growth and decay models (exponential growth, logistic growth), spread of diseases, modelling predator-prey dynamics. • Radioactive decay, reaction rates (first order and second-order reactions), concentration changes over time.	– Real-life modelling projects: Students are given open-ended tasks where students choose a real-world scenario (population, chemical reaction, circuit response, or cooling object), formulate a differential equation model, solve it, and interpret the results. Students present findings with graphs and tables in a whole class discussion.
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			• Modelling economic growth, demand-supply dynamics, depreciation of assets.	
<i>Recognize particular and general solutions of differential equations.</i>				
<i>Assessment criteria: Students solve first- and second-order differential equations accurately using appropriate methods, apply them in modelling real-life situations, and demonstrate logical reasoning, accuracy, and confidence in presenting solutions.</i>				
<i>Teaching and learning materials: Graphing calculators, GeoGebra, similar software, graph paper, and any AI.</i>				

Topic area: Linear Algebra			Sub-topic Area: Vectors	
S.6 MATHEMATICS	Unit 6: intersection and sum of subspaces.			Number of periods: 7
Key unit competence: Relate the sum and the intersection of subspaces of a vector space by the dimension formula				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe the intersection and sum of subspaces. – State Grassmann's dimension formula. – Identify whether a given subset is an intersection or sum of specific subspaces.	– Compute the intersection and sum of given subspaces. – Determine the dimensions of the intersection and sum of subspaces.	– Appreciate the logical structure of Grassmann's formula. – Value precision in defining mathematical objects and performing calculations.	– Definition of subspaces – Intersection of subspaces – Dimensions of intersection of subspaces – Sum of subspaces – Dimension of sum of subspaces – Grassmann's formula of dimension for subspaces	– Geometric Visualization: In small groups, students use 3D models or dynamic geometry software (like GeoGebra) to visualize the intersection and sum of subspaces in $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ (e.g., two planes intersecting in a line, a plane and a line summing to the whole space).

	– Apply Grassmann's formula to verify calculations and solve for unknown dimensions. – Analyse the relationship between intersecting and summing subspaces through their dimensions.	– Collaborate effectively with peers to solve complex problems.		– Students present their findings to the class. – Each pair discusses whether the given sets of vectors is subspaces e.g: $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ and $W = \{(x, y, z) \in \mathbb{R}^3 : x - y = 0\}$. – Determine the intersection $U \cap W$ and represent it graphically on coordinate axes to visualize how two subspaces can intersect. – Formula Derivation Challenge: Through a guided discovery activity, students derive Grassmann's formula for two subspaces using basis vectors for each subspace and their intersection. – Use two subspaces to determine its sum, intersection dimension and verify Grassmann's Formula
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				e.g: $U = \text{span}\{(1, 0, 0), (0, 1, 0)\}$ $W = \text{span}\{(1, 0, 0), (0, 0, 1)\}$ Verify whether $\dim(U + W) = \dim(U) + \dim(W)$ $-\dim(U \cap W)$
Link to other subjects: Physics: Concept of vector spaces in mechanics (e.g., forces in a plane).				
Assessment criteria: The learner can accurately compute the intersection and sum of given subspaces, correctly determine their dimensions, and reliably apply Grassmann's formula to solve problems and verify results.				
Teaching and learning materials: Markers, GeoGebra or other dynamic geometry software or any AI, physical models (e.g., planes made of cardboard), worksheets with problem sets, textbooks, and projector.				

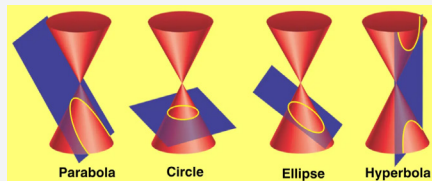
Topic area: Linear algebra			Sub-topic area: Linear Transformations	
S6 Mathematics		Unit 7: Transformation Of Matrices		Number of periods:21
Key Unit Competence: Applying matrix transformations into echelon or diagonal form in solving simultaneous linear equations and computing powers of matrices.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain the process of elementary row and column operations. – Recall the concept of kernel, range, eigenvalues, eigenvectors, and matrix diagonalization. – Explain the applications of echelon form and diagonalization	– Perform row and column operations to transform matrices to echelon form. – Compute kernel, range, rank, eigenvalues, and eigenvectors. – Diagonalize matrices and calculate matrix powers.	– Demonstrate accuracy and precision in computations. – Develop logical reasoning and problem-solving skills in echelon matrix – Show curiosity and interest in real-life applications of row operation matrix and diagonalization	– Elementary row operations: Elementary row/ column operations on matrices, Echelon matrix – Kernel and Range – Diagonalisation of matrices: Eigen values and Eigen vectors of a matrix, diagonalization of matrices. – Application of echelon form/row operation:	– Pair-share: Practice elementary row operations on small matrices and compare results with a partner. – Problem-solving: Transform given matrices to echelon form and determine kernel, range, and rank e.g: $\begin{pmatrix} \ddot{u} \\ \ddot{u} \\ \ddot{u} \end{pmatrix}$

	– Solve systems of linear equations using matrix transformations. – Apply matrix operations to real-life problems.		• To find rank of matrix • To find inverse matrix • To solve system of linear equations – Application of diagonalization: Power of a matrix.	– Group activity: Find inverses of given matrices using row operation collaboratively $\text{eg: } \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{pmatrix}$ – Guided practice: Diagonalize matrices and calculate powers of matrices using diagonal form $\text{e.g: } M = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}, \text{ find } M^5.$
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Link to other subjects: **Physics** (rotations); **Economics** (Solving systems of linear equations in modelling and optimization problems).

Assessment criteria: Students accurately transform matrices into echelon or diagonal form, compute kernel, range, rank, inverses, eigenvalues, eigenvectors and apply them to solve system of linear equation and power of matrix.

Teaching and learning materials: Manila paper, calculators for numerical computations.

Topic area: Geometry			Sub-topic area: Plane Geometry	
S6 Mathematics		Unit 8: Conics		Number of periods: 28
Key Unit Competence: Analyze the properties and equations of conic sections by applying them to solve problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe conic sections and classify them (circle, ellipse, parabola, hyperbola). – State the geometric definition, standard equations, and key features (vertex, focus, directrix, axis) of a parabola.	– Classify a given conic section from its general equation. – Sketch the graph of a parabola from its equation, labelling all key features. Derive the equation from given characteristics.	– Develop a systematic approach to analysing and graphing more complex curves. – Recognize the utility of different coordinate systems for solving different problems.	– Generalities on conic sections – Definition and classification of conics – Parabola: Definition, equations, characteristics and graph, tangent and normal line. – Ellipse: Definition, equations, characteristics and graph, tangent and normal line	– Teacher guides students to use physical models or geometry software to slice a cone and discover the different conic sections:  Then, students classify the curves based on the angle of the cut.

– State the geometric definition, standard equations, and key features (foci, vertices, axes) of an ellipse. – State the geometric definition, standard equations, and key features (foci, vertices, asymptotes) of a hyperbola. – Describe the polar coordinate system and the polar equations of conics. – Identify applications of conic sections in various fields.	– Sketch the graph of an ellipse from its equation. Derive its equation from given features. Apply the reflective property to solve problems. – Sketch the graph of a hyperbola, including its asymptotes. Derive its equation from given features.	– Foster creativity and appreciate the role of conics in technology and science.	– Hyperbola: Definition, equations, characteristics and graph, tangent and normal line. – Polar form: Definition of polar coordinates, Polar equation of a conic, straight line and circle. – Applications for Conics in real life and other sciences.	– Given a parabola's equation, students plot its graph, identify features, and derive the equation of the tangent and normal at a given point. Students solve real-life problems (e.g., projectile motion, satellite dishes). – Students construct an ellipse using pins and a string to understand its geometric definition. Given an equation, students graph the ellipse and find the equations of tangent and normal lines. – In groups, use graphing software to explore the effect of parameters on a hyperbola's shape and asymptotes. – Solve problems involving the equations of tangents and normal lines to a conic. – Use polar equations to plot the graphs of conics.
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	– Convert equations between polar and Cartesian forms. Identify the type of conic from its polar equation using eccentricity. – Model simple real-life situations using the properties of conics.			– Analyse how the eccentricity in the polar form of the conic $r = \frac{ed}{1 + e \cos \theta}$ determines the – shape of the conic section depending on the value of e – Conduct a group research project on real-world applications (e.g., ellipses in orbits, parabolas in headlights, hyperbolas in navigation) and present findings to the class.
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Link to other subjects: **Physics** (Conic paths in projectile motion and planetary orbits); **Geography** (Use of ellipses and circles in mapping and geolocation).

Assessment criteria: Students determine the characteristics and the graph of a conic given by its Cartesian, parametric or polar equation and find the Cartesian, parametric and polar equations of a conic from its characteristics Correctly.

Teaching and learning materials: GeoGebra, graph papers, rulers, compasses, and protractors; Models of cones and strings for hands-on demonstrations.

Topic area: Statistics and probability			Sub-topic area: Probability	
S6 Mathematics		Unit 9: Random Variables		Number of periods: 21
Key Unit Competence: Calculate the parameters of a random variable (discrete or continuous) including binomial and the Poisson distributions.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe discrete and continuous random variables. – Recall probability density functions (PDF) and cumulative distribution functions (CDF). – Explain expected value, variance, and standard deviation.	– Compute probabilities, expected value, variance, and standard deviation for discrete and continuous random variables. – Apply binomial and Poisson formulas to practical problems.	– Develop logical reasoning and critical thinking in probability problems. – Appreciate the importance of probability distributions in modelling uncertainty.	Discrete and finite random variable: – Introduction to random variable – Probability density function of a discrete random variable – Cumulative distribution of discrete random variable – Expected value, variance and standard deviation of a discrete random variable – Binomial distribution – Expected value of a binomial distribution	– Pair-Share: Identify discrete and continuous random variables in real-life examples (e.g., dice rolls, waiting times). – Problem-Solving: in pairs, calculate probabilities and cumulative probabilities for a given discrete distribution. – Guided practice: In groups, compute expected value, variance, and standard deviation of a discrete random variable: determine probabilities for binomial distribution $P(X=k)$ for $n=5$ and $p = 0.6$.

– State the properties and applications of binomial and Poisson distributions.	– Construct and interpret PDF and CDF graphs. – Analyse and compare discrete and continuous random variables in real-life contexts.	– Demonstrate accuracy and precision in probability calculations	– Variance and standard deviation of a binomial distribution – Poisson distribution Continuous random variables: – Probability density function – Cumulative distribution – Expected value, variance, and standard deviation. – Main types of continuous distributions (normal, continuous uniform and exponential distributions)	– Investigation or application: Use Poisson distribution to model events like number of arrivals in an hour. – Individual work: Calculate probability density functions (PDF) and cumulative distribution functions (CDF), mean, variance, and standard deviation of a continuous random variable (e.g: uniform or exponential).
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Link to other subjects: **Biology** (Population studies, genetics, or random sampling of organisms); **Economics** (Forecasting demand, modelling market events, or customer arrivals).

Assessment criteria: Students accurately compute probabilities, expected values, variances, and standard deviations of random variables, apply binomial and Poisson distributions, interpret results and construct PDF and CDF graphs.

Teaching and learning materials: Software tools (GeoGebra, Excel, or AI).

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ANNEX 1: NUMBER OF PERIODS PER SUBJECT FOR MATH AND SCIENCES PATHWAY

a) Mathematics and Science Stream I

Subject	Number of periods		
I. Major Core subjects: Examinable at National level	(Period= 40 minutes)		
	S4	S5	S6
1. Mathematics	7	7	7
2. Physics	7	7	7
3. Chemistry	7	7	7
4. Biology	7	7	7
II. Minor Core Subjects: Examinable at National level			
5. English	3	3	3
6. French	3	3	3
7. Kinyarwanda	3	3	3
Sub-total 1	37	37	37
III. General Subjects examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and Religious studies	3	3	3

11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300

b) Mathematics and Science Stream 2

Subject	Number of periods		
I. Major Core Subjects: Examinable at National Level	(Period= 40 minutes)		
	S4	S5	S6
1. Mathematics	7	7	7
2. Economics	7	7	7
3. Geography	7	7	7
4. Physics	7	7	7
II. Minor Core Subjects: Examinable at National Level			
5. English	3	3	3
6. French	3	3	3
7. Kinyarwanda	3	3	3
Sub-total 1	37	37	37

III. General Subjects examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and religious studies	3	3	3
11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300